
Understanding Task Vectors in In-Context Learning: Emergence, Functionality, and Limitations

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Abstract

Task vectors offer a compelling mechanism for accelerating inference in in-context learning (ICL) by distilling task-specific information into a single, reusable representation. Despite their empirical success, the underlying principles governing their emergence and functionality remain unclear. This work proposes the *Linear Combination Conjecture*, positing that task vectors act as single in-context demonstrations formed through linear combinations of the original ones. We provide both theoretical and empirical support for this conjecture. First, we show that task vectors naturally emerge in linear transformers trained on triplet-formatted prompts through loss landscape analysis. Next, we predict the failure of task vectors on representing high-rank mappings and confirm this on practical LLMs. Our findings are further validated through saliency analyses and parameter visualization, suggesting an enhancement of task vectors by injecting multiple ones into few-shot prompts. Together, our results advance the understanding of task vectors and shed light on the mechanisms underlying ICL in transformer-based models.

1 Introduction

In-context learning (ICL) is a core capability of large language models (LLMs), allowing them to perform new tasks without parameter updates by conditioning on a few input-output examples in the prompt [2]. Unlike traditional training, ICL relies on attention-based mechanisms to infer task structure directly from context. This surprising generalization ability has led to growing interest in uncovering the principles of learning purely from contextual examples [21, 3, 4, 15, 5].

A recent work investigates the task vector method [7] (concurrent works include function vectors [16] and in-context vectors [13]), a technique that distills underlying task information from ICL demonstrations into a single vector. Typically, ICL prompts are structured as sequences of triplets, each encoding a semantic mapping, in addition to a query at the end (e.g., “*hot* → *cold*, *up* → *down*, *day* → *night*, *dark* → ”). Task vectors are then extracted from the hidden states of the last (→) token. Once obtained, these vectors can be injected into the same position in new prompts (e.g., “*big* → ”), enabling the model to generalize to unseen inputs in a zero-shot fashion.

Task vectors have been shown to naturally emerge even in small transformer models trained from scratch on synthetic data [24], suggesting that their formation is a general property of attention-based architectures. Recent studies further demonstrate that task vectors can be enhanced by aggregating hidden states across multiple layers and multiple arrow tokens [12]. Beyond language models, task vectors are also found effective in large-scale visual [8] and multi-modal [9] models.

Despite their empirical effectiveness, the underlying mechanism of task vectors, especially how they emerge, function, and encode task information, remains poorly understood. This paper takes a step toward unveiling the principles behind it by introducing the following conjecture:

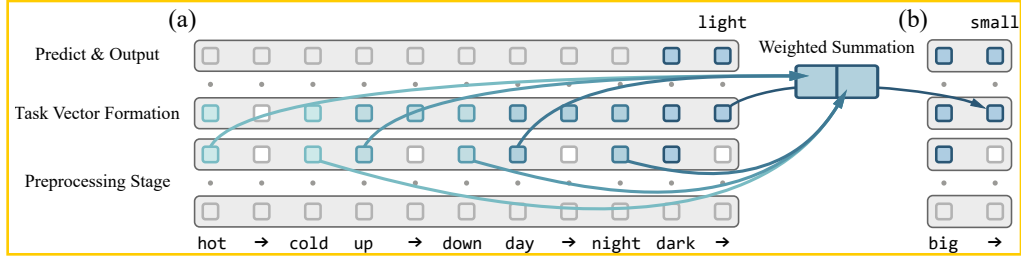


Figure 1: Overview of task vector and our main conjecture. (a) Task vector emerges during ICL as a linear combination of preceding in-context demonstrations. (b) It can then be injected into zero-shot prompts and functions as a single, representative demonstration, facilitating efficient prediction.

Linear Combination Conjecture

The injected task vector functions as a single in-context demonstration, formed through a linear combination of the original demonstrations (hidden states).

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Figure 1 provides an intuitive illustration for our conjecture. In the following sections, we validate this conjecture through various empirical and theoretical perspectives. These analyses comprehensively explain how task vectors naturally emerge within attention-based model architectures, effectively encode task-related information, and facilitate inference in zero-shot prompts. Our work advances the understanding of the underlying mechanisms behind ICL, clarifying both the efficacy and limitations of task vectors in transformer-based LLMs. The highlights of this paper are as follows:

- **Theoretical Justification in Linear Transformers:** We theoretically characterize the critical points of linear-attention transformers and demonstrate how they solve random linear regression tasks through embedding concatenation and gradient descent. With a triplet-formatted input prompt structure, task vectors naturally emerge at arrow tokens as linear combinations of the in-context demonstrations. These vectors serve as redundancy against information loss induced by dropout, thereby improving robustness. Empirically, the learned linear model parameters closely align with the predicted structure and successfully replicate the task vector mechanism.
- **Empirical Verification in Practical LLMs:** We visualize the information flow in LLMs with saliency analysis and observe patterns consistent with linear models, suggesting they share similar underlying mechanisms. According to our conjecture, inference with task vectors is analogous to 1-shot ICL, which is inherently limited to rank-one meta-predictors under the gradient descent perspective. To validate this, we introduce a series of bijection tasks that are provably unsolvable by rank-one predictors, and empirically confirm this failure in real-world transformers. Building on these insights, we enhance the standard task vector method by injecting multiple vectors into few-shot prompts, resulting in consistent performance gains across a range of ICL tasks.

2 Setting: Random Linear Regression with Linear-Attention Transformers

Notations: We write $[n] = \{1, \dots, n\}$. The Hadamard product is denoted by $\odot$, and the Kronecker product by $\otimes$. The identity matrix of dimension n is denoted by I_n , while 0_n and $0_{m \times n}$ represent zero vectors or matrices of the corresponding dimensions. Subscripts are omitted when the dimensions are clear from context. We define $\mathcal{M}(M) = \{\Lambda \in \mathbb{R}^{\dim(M)} \mid \Lambda = M \circ A, A \in \mathbb{R}^{\dim(M)}\}$ as the set of masked matrices induced by the binary mask M . For a general matrix A , the element at the i -th row and j -th column is denoted by $A_{i,j}$, and the sub-block from rows i to k and columns j to l is denoted by $A_{i:k,j:l}$. $\text{diag}(A_1, \dots, A_n)$ represents the block-diagonal matrix constructed by $\{A_i\}_{i=1}^n$.

Random Linear Regression: Following the settings in literature [6, 17, 1, 20], we consider training linear transformers on random instances of linear regression. Let $\{x_i\}_{i=1}^{n+1}$, where $x_i \in \mathbb{R}^d$, denote covariates drawn i.i.d. from distribution P_x , and let $\{w_i\}_{i=1}^d$, where $w_i \in \mathbb{R}^d$, denote coefficients drawn i.i.d. from distribution P_w . Define the coefficient matrix as $W = [w_1 \dots w_d]^\top \in \mathbb{R}^{d \times d}$. The responses are then generated as $y_i = Wx_i$ for $i \in [n+1]$. We denote by $X, Y \in \mathbb{R}^{d \times n}$ the matrices whose columns are x_i and y_i , respectively, for $i \in [n]$. The query covariate and response are denoted by $x_{\text{test}} = x_{n+1}$ and $y_{\text{test}} = y_{n+1}$ respectively.

Linear Self Attention Transformer: Following prior works [17, 1, 20], we consider transformers composed of linear self-attention layers. Let $Z_0 \in \mathbb{R}^{2d \times d_p}$ denote the input matrix constructed from X, Y and x_{test} but excluding y_{test} , where d_p denotes the number of tokens. The transformer is defined by stacking L attention blocks with skip connections, where the l -th layer is expressed as:

$$Z_l = Z_{l-1} + \frac{1}{n} \text{Attn}_{V_l, Q_l}(Z_{l-1}), \quad \text{Attn}_{V, Q}(Z) = V Z M (Z^\top Q Z). \quad (1)$$

Here, the trainable parameters are $\{V_l, Q_l\}_{l=1}^L$, where $V_l \in \mathbb{R}^{2d \times 2d}$ represents a reparameterization of the projection and value matrices, and $Q_l \in \mathbb{R}^{2d \times 2d}$ denotes the query and key matrices. Following the work [1], we adopt a masking matrix $M = \text{diag}(I_{d_p-1}, 0)$ to prevent attention from earlier tokens to the final one. The output of the transformer is defined as $\text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^L) = (Z_L)_{(d+1:2d), d_p}$ (i.e., the latter half of the last column). This definition aligns with the structure of the input Z_0 , which will be further discussed in subsequent sections. During training, the parameters are optimized to minimize the expected ICL risk over random linear regression instances:

$$\mathcal{L}(\{V_l, Q_l\}_{l=1}^L) = \mathbb{E}_{Z_0, W} \|\text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^L) + W x_{\text{test}}\|_2^2. \quad (2)$$

3 Emergence of Task Vectors in Linear-Attention Transformers

Firstly, we present theoretical evidence indicating that task vectors naturally arise even in simple linear transformers. Specifically, we analyze the loss landscape of the in-context risk, focusing on the properties of its critical points. As a startup, recall the standard linear regression setup [1, 20], where the (x_i, y_i) pairs for each demonstration are concatenated to form the input prompt:

$$Z_0 = \begin{bmatrix} X & x_{\text{test}} \\ Y & 0 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n & x_{\text{test}} \\ y_1 & y_2 & \cdots & y_n & 0 \end{bmatrix} \in \mathbb{R}^{2d \times (n+1)}. \quad (3)$$

According to existing analyses [1, 25, 14], each attention layer in this setting performs one step of gradient descent on the coefficient matrix W . Specifically, the theoretically optimal single-layer (possibly nonlinear) attention [10] implements the following predictive function [1] when the covariates are drawn from $P_x = \mathcal{N}(0, I_d)$, by selecting $V_1 \propto \text{diag}(0_{d \times d}, I_d)$ and $Q_1 \propto \text{diag}(I_d, 0_{d \times d})$:

$$\text{TF}(Z_0; (V_1, Q_1)) = -\frac{1}{n} Y \sigma(X)^\top \sigma(x_{\text{test}}), \quad \text{where } \sigma : \mathbb{R}^d \mapsto \mathbb{R}^r \text{ is a kernel function.} \quad (4)$$

Here, we abbreviate $[\sigma(x_1) \cdots \sigma(x_n)]$ as $\sigma(X)$. Equation (4) employs $W' \propto Y \sigma(X)^\top$ as an estimate of coefficient matrix W , yielding prediction $\hat{y}_{\text{test}} = W' \sigma(x_{\text{test}})$. In this paper, we consider alternative settings more reflective of practical scenarios, where x_i and y_i are separated as distinct tokens. As noted [26], such separation necessitates the usage of position encodings for bi-directional attention. Following prior analysis [11], we assume that position encodings are appended to the input tokens, and reformulate the layer-wise update rule of self-attention as:

$$\text{Attn}_{V, Q}(Z) = V Z M \begin{bmatrix} Z^\top & P^\top \end{bmatrix} Q \begin{bmatrix} Z \\ P \end{bmatrix}, \quad \text{where } P \in \mathbb{R}^{d_p \times d_p}. \quad (5)$$

For analytical tractability, we take $P = I_{d_p}$ as one-hot position encodings. Inspired by the parameter structure in [1] and eq. (4), we further impose the following constraints on the trainable parameters:

$$V_l = \text{diag}(A_l, B_l), \quad Q_l = \text{diag}(C_l, 0_{d \times d}, D_l), \quad \text{where } A_l, B_l, C_l \in \mathbb{R}^{d \times d}, D_l \in \mathbb{R}^{d_p \times d_p}. \quad (6)$$

These parameterizations ensure that the projection and attention operations act independently on the covariate, response, and positional components of the input. This structural decoupling is essential for understanding how the transformer identifies the dependency between each (x_i, y_i) pair and revealing the actual optimization algorithm being executed by the model. The proofs for the main theoretical results in this paper are available in Appendix B.

3.1 Warm-up: Learning with Pairwise Demonstrations

We begin by analyzing the optimization of linear transformers on pairwise demonstrations. Following previous approach [6, 19, 22], we decompose each demonstration in eq. (3) into a pair of tokens

$$Z_0^i = \begin{bmatrix} x_i & 0 \\ 0 & y_i \end{bmatrix} \in \mathbb{R}^{2d \times 2}$$

to better reflect the practical ICL prompt structure:

$$Z_0 = \begin{bmatrix} Z_0^1 & \cdots & Z_0^n & Z_0^{\text{test}} \end{bmatrix} = \begin{bmatrix} x_1 & 0 & \cdots & x_n & 0 & x_{\text{test}} & 0 \\ 0 & y_1 & \cdots & 0 & y_n & 0 & 0 \end{bmatrix} \in \mathbb{R}^{(2d) \times (2n+2)}. \quad (7)$$

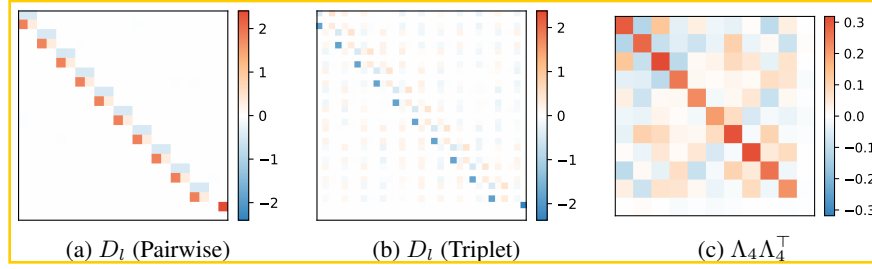


Figure 2: Visualization of learned D_l weights. (a) Pairwise demonstrations yield a block-diagonal structure aligned with Theorem 1. (b) Triplet demonstrations yield a richer structure aligned with Theorem 2. (c) The learned matrix Λ_4 has nearly orthonormal rows as suggested by Proposition 3.

The following theorem suggests that certain critical points of the in-context risk effectively solve the regression problem by first concatenating each pair of (x_i, y_i) into the same tokens, and then executing a variant of the gradient descent algorithm to compute the prediction. To simplify notation, we denote $A = \{A_l\}_{l=1}^L$ (similarly for B, C , and D) and present:

Theorem 1 (Critical Points; Pairwise Demonstrations). Assume $P_x = \mathcal{N}(0, \Sigma)$, $P_y = \mathcal{N}(0, \Sigma^{-1})$ with some $\Sigma \in \mathbb{R}^{d \times d}$ satisfying $\Sigma \succ 0$. Define $\mathcal{S}_I, \mathcal{S}_\Sigma \subset \mathbb{R}^{d \times d}$ and $\mathcal{S}_P \subset \mathbb{R}^{d_p \times d_p}$ as

$$\mathcal{S}_I = \{\lambda I_d \mid \lambda \in \mathbb{R}\}, \quad \mathcal{S}_\Sigma = \{\lambda \Sigma^{-1} \mid \lambda \in \mathbb{R}\}, \quad \mathcal{S}_P = \{\text{diag}(I_n \otimes \Lambda_1, \Lambda_2) \mid \Lambda_1, \Lambda_2 \in \mathbb{R}^{2 \times 2}\}.$$

Consider optimizing an L -layer linear transformer with pairwise demonstrations and parameter configuration given in eq. (6), we then have

$$\inf_{A, B \in \mathcal{S}_I^L, C \in \mathcal{S}_\Sigma^L, D \in \mathcal{S}_P^L} \sum_{H \in A \cup B \cup C \cup D} \|\nabla_H \mathcal{L}(\{V_l, Q_l\}_{l=1}^L)\|_F^2 = 0.$$

To understand the behavior of these critical points within a self-attention layer, we fix $\Sigma = I_d$ and take $A_l, B_l = I_d, C_l = -\lambda I_d$, and $D_l = \text{diag}(I_n \otimes \Lambda_1, \Lambda_2)$. Let the first and last d rows of Z_l be denoted by X_l and Y_l , respectively. Under these settings, the update rule of each layer becomes:

$$Z_l = Z_{l-1} - \lambda Z_{l-1} M X_{l-1}^\top X_{l-1} + [Z_{l-1}^1 \Lambda_1 \quad \dots \quad Z_{l-1}^n \Lambda_1 \quad Z_{l-1}^{\text{test}} \text{diag}(1, 0) \Lambda_2]. \quad (8)$$

The above update can be decomposed into the following two distinct components:

- **Gradient Descent:** The first component, $Z_l \leftarrow Z_{l-1} - \lambda Z_{l-1} M X_{l-1}^\top X_{l-1}$, implements the GD++ algorithm [17]. This variant enhances convergence speed over standard gradient descent by improving the condition number of the Gram matrix $X_{l-1}^\top X_{l-1}$. Notably, this operation modifies only X_l but not Y_l for the first layer, as implied by the structure of Q_l (eq. (6)).
- **Embedding Concatenation:** The second component, $Z_l^i \leftarrow Z_{l-1}^i + Z_{l-1}^i \Lambda_1$ for $i \in [n]$, mixes each pair of (x_i, y_i) tokens. Given that x_i and y_i tokens are initially linearly separable as in our formulation, this operation concatenates each (x_i, y_i) pair, thereby transforming pairwise demonstrations into the original single-token format. For the query token Z_l^{test} , this operation copies x_{test} into the final token, reconstructing the structure in eq. (3), where each non-final token directly concatenates (x_i, y_i) of a demonstration, and the final token contains only x_{test} .

In summary, our analysis reveals that for pairwise demonstrations, the first attention layer leverages position encodings to distinguish between covariate and response tokens, subsequently concatenating them to form a single-token prompt structure. The remaining layers then apply the GD++ algorithm, mirroring the learning dynamics on single-token demonstrations. As a result, an L -layer linear transformer allocates one layer for embedding concatenation and utilizes the remaining $L - 1$ layers to perform gradient descent. In Figure 2a, we visualize the learned D_l weights under the setting of Theorem 1, and observe that they closely match the critical point structure of $\mathcal{S}_P$.

3.2 Emergence of Task Vectors with Triplet Demonstrations

Next, to better reflect the prompt structure of practical ICL, we insert additional zero tokens between each pair of (x_i, y_i) to simulate the arrow ($\rightarrow$) tokens. This reformulates each demonstration as a triplet $(x_i, \rightarrow, y_i)$, enabling us to analyze the critical points with these triplet demonstrations:

$$Z_0 = \begin{bmatrix} x_1 & 0 & 0 & \dots & x_n & 0 & 0 & x_{\text{test}} & 0 & 0 \\ 0 & 0 & y_1 & \dots & 0 & 0 & y_n & 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{(2d) \times (3n+3)}. \quad (9)$$

144 **Theorem 2 (Critical Points: Triplet Demonstrations).** Assume $P_x = \mathcal{N}(0, \Sigma)$, $P_w = \mathcal{N}(0, \Sigma^{-1})$
 145 with some $\Sigma \in \mathbb{R}^{d \times d}$ satisfying $\Sigma \succ 0$. Define $\mathcal{S}_I, \mathcal{S}_\Sigma \subset \mathbb{R}^{d \times d}$ and $\mathcal{S}_P \subset \mathbb{R}^{d_p \times d_p}$ as

$$\begin{aligned} \mathcal{S}_I &= \{\lambda I_d \mid \lambda \in \mathbb{R}\}, \quad \mathcal{S}_\Sigma = \{\lambda \Sigma^{-1} \mid \lambda \in \mathbb{R}\}, \\ \mathcal{S}_P &= \left\{ \text{diag}(I_n \otimes \Lambda_1, \Lambda_2) + I_{n+1} \otimes \Lambda_3 + \Lambda_4 \otimes \Lambda_5 \mid \right. \\ &\quad \left. \Lambda_1, \Lambda_2 \in \mathcal{M}\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}\right), \Lambda_3 \in \mathcal{M}\left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}\right), \Lambda_4 \in \mathbb{R}^{(n+1) \times (n+1)}, \Lambda_5 \in \mathcal{M}\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}\right) \right\}. \end{aligned}$$

146 Consider optimizing an L -layer linear transformer with triplet demonstrations and parameter config-
 147 uration given in eq. (6), we then have

$$\inf_{A, B \in \mathcal{S}_I^L, C \in \mathcal{S}_\Sigma^L, D \in \mathcal{S}_P^L} \sum_{H \in A \cup B \cup C \cup D} \|\nabla_H \mathcal{L}(\{V_l, Q_l\}_{l=1}^L)\|_F^2 = 0.$$

148 To analyze the behavior of each attention layer, we note that the critical points for the matrices A_i ,
 149 B_i , and C_i remain consistent with Theorem 1, thereby implementing the GD++ algorithm. For the
 150 matrix D_i , we decompose its structure into three distinct components:

- 151 • **Embedding Concatenation:** The first component, $\text{diag}(I_n \otimes \Lambda_1, \Lambda_2)$, mixes each pair of (x_i, y_i)
 152 tokens, effectively concatenating them — analogous to the operation analyzed in the previous
 153 section. This converts all non-arrow tokens into single-token demonstrations.
- 154 • **Self Magnification:** The second component, $I_{n+1} \otimes \Lambda_3$, scales the embeddings corresponding to
 155 each arrow ($\rightarrow$) token by a fixed constant and adds them back to themselves.
- 156 • **Task Vector Formation:** The third component, $\Lambda_4 \otimes \Lambda_5$, performs a weighted summation across
 157 all demonstrations in the prompt. This operation is central to the emergence of task vectors. Let
 158 $[\beta_1 \ \dots \ \beta_{n+1}] \in \mathbb{R}^{n \times (n+1)}$ denote the first n rows of Λ_4 (we will soon show that the last row
 159 of Λ_4 converges to zero), the first self-attention layer then outputs $n+1$ linear combinations of
 160 the demonstrations as the hidden states for the arrow tokens, expressed as $z_{tv}^i = [\alpha_1 X \beta_i]$ for
 161 $i \in [n+1]$ where $\alpha_1, \alpha_2 \in \mathbb{R}$ are the two non-zero entries of Λ_5 . These vectors can then be
 162 injected into zero-shot prompts and function as single-token demonstrations.

163 This mechanism provides strong theoretical evidence for our linear combination conjecture, demon-
 164 strating that **task vectors naturally emerge from the optimization dynamics of linear-attention**
 165 **transformers operating on triplet-formatted prompts.** Notably, the structure of $\mathcal{S}_P$ closely aligns
 166 with our visualization of D_i in Figure 2b, confirming our theoretical analysis. We now further
 167 investigate the structure of the weight matrix Λ_4 , and present the following result:

168 **Proposition 3 (Optimal Task Vector Weights).** Assume $P_x, P_w = \mathcal{N}(0, I_d)$. Consider optimizing
 169 a 2-layer linear-attention transformer with triplet demonstrations and parameter configuration given
 170 in eq. (6), and assume $C_1 = 0$. Let

$$D_1 = \text{diag}(I_n \otimes \Lambda_1, \Lambda_2) + I_{n+1} \otimes \Lambda_3 + \Lambda_4 \otimes \Lambda_5 \in \mathcal{S}_P$$

171 be any minimizer of the in-context risk $\mathcal{L}(\{V_l, Q_l\}_{l=1}^L)$, we then have $\Lambda_4 \in \mathcal{S}_U$, where

$$\mathcal{S}_U = \{\Lambda \mid \Lambda \Lambda^\top = \lambda \text{diag}(I_n, 0), \lambda \in \mathbb{R}\}.$$

172 This result suggests that the optimal Λ_4 weight matrix satisfies two key properties: (1) the last row
 173 is zero, and (2) the first n rows are mutually orthonormal. These conditions imply that the learned
 174 weight vectors $\beta_1, \dots, \beta_{n+1}$ are likely to be distinct. Therefore, the $n+1$ task vectors produce
 175 diverse linear combinations of the demonstrations, thereby enriching the representation within the
 176 input prompt. This implication is verified in Figure 2c. While task vectors are typically extracted
 177 from the final arrow ($\rightarrow$) token in standard usage, here we consider all arrow tokens as task vectors
 178 as bi-directional attention allows each to aggregate information from the full prompt.

179 4 Validating the Linear Combination Conjecture on Bijection Tasks

180 We then present an empirical observation that supports our conjecture. Consider the setting where
 181 task vectors are injected into zero-shot prompts. Based on our prior analysis, the injected task vector
 182 z_{tv} is formed as a linear combination of the original demonstrations. As a result, we show that the
 183 injected prompt reconstructs the single-token structure in eq. (3) with only 1 demonstration:

$$Z_0 = [z_{\text{test}} \quad z_{\text{tv}} \quad 0] = \begin{bmatrix} x_{\text{test}} & x_{\text{tv}} & 0 \\ 0 & y_{\text{tv}} & 0 \end{bmatrix} = \begin{bmatrix} x_{\text{test}} & X\beta & 0 \\ 0 & Y\beta & 0 \end{bmatrix} \in \mathbb{R}^{2d \times 3}, \quad (10)$$

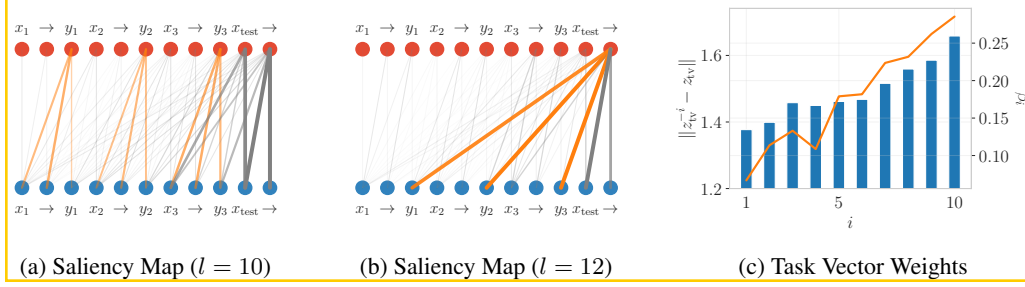


Figure 3: Visualization of saliency matrices as bipartite graphs between layer l (●) and $l+1$ (●), where edge widths indicate saliency magnitude; and variations in the extracted task vector after perturbing the i -th demonstration (■), alongside the predicted weights (—) obtained by optimizing Proposition 6. (a) Each y_i token is primarily attending to its corresponding (x_i, y_i) pair, reflecting embedding concatenation. (b) The final ($\rightarrow$) token attends broadly to all y_i tokens, indicating task vector formation — this occurs just before the optimal injection layer $l = 13$. (c) The predicted task vector weights closely match the trend of empirical results, validating our theoretical model.

where the weight vector $\beta \in \mathbb{R}^n$ comes from the last column of Λ_4 (Theorem 2). After the first layer, the Λ_2 matrix of $\mathcal{S}_E$ moves x_{test} to the last token, reducing the prompt to a single-shot, single-token demonstration. According to the optimal single-layer transformer (eq. (4)), the estimated coefficient matrix is now $W' = Y\beta(X\beta)^T$, which is rank-one. Therefore, if our main conjecture holds, task vectors will be inherently limited in their expressiveness: *they can only realize rank-one coefficient matrices*. This implication also naturally extends to multi-layer transformers.

While our analysis is conducted on linear-attention transformers, we demonstrate that similar learning patterns also emerge within practical LLMs. Specifically, we visualize the layer-wise information flow between tokens using saliency maps [18], where the saliency score for each attention matrix is computed as $S(A_l) = \sum_i |A_{l,h} \cdot \partial \mathcal{L} / \partial A_{l,h}|$. $A_{l,h}$ denotes the attention matrix of the h -th head at layer l and $\mathcal{L}$ is the ICL loss (i.e., the cross-entropy loss for predicting y_{test}). As demonstrated in Figures 3a and 3b, the saliency maps reveal certain patterns matching the ones of embedding concatenation and weighted summation. Importantly, the latter occurs immediately before the optimal task vector injection layer. This suggests that real-world models implement a similar algorithm to solve ICL tasks and, consequently, inherit the same expressiveness limitation.

To verify this, we construct a specialized class of ICL tasks, named bijection tasks. Specifically, given a bijective mapping from domain $\mathcal{X}$ to codomain $\mathcal{Y}$, one can combine it with its inverse mapping to form a new task that maps $\mathcal{X} \cup \mathcal{Y}$ onto itself. For instance, combining the "to uppercase" task with its inverse "to lowercase" yields a bijection task that maps each letter to its opposite case, and a valid ICL prompt takes the form: " $a \rightarrow A, B \rightarrow b, c \rightarrow C, D \rightarrow d$ ". Note that this differs from task superposition [23], as each input corresponds to a unique, well-defined output. We then establish a key limitation of rank-one coefficient matrices in addressing such tasks:

Proposition 4. Let $x, y \in \mathbb{R}^d$ be non-zero vectors. Then the following are equivalent: (1) There exists a rank-one matrix $W \in \mathbb{R}^{d \times d}$ such that $y = Wx$ and $x = Wy$; (2) $x = y$ or $x = -y$.

This result highlights that rank-one coefficient matrices cannot solve general bijection tasks, and are restricted to only the identity mapping ($x = y$) or the negation mapping ($x = -y$). We further verify this implication in real-world LLMs: as summarized in Table 1, both ICL and the task vector method perform well on the original tasks and their inverses. Nevertheless, for the bijection tasks, while ICL preserves performance in many cases, the task vector method consistently fails, confusing examples from the two domains and yielding near-random predictions (50%). For instance, in the "to uppercase" task, task vectors can predict the correct letter but fail to distinguish between uppercase and lowercase. The only notable exceptions are the copy task (corresponding to the $x = y$ case in Proposition 4) and the antonym task (corresponding to $x = -y$).

Together, these findings empirically validate our conjecture: **the task vector approach, which is restricted to rank-one coefficient matrices, cannot solve general bijection tasks**. While a variety of ICL tasks have been explored to assess the capabilities of task vectors [7, 16, 12], the fundamental limitation of task vectors in addressing these bijection tasks has not been previously identified.

Table 1: Comparison of the accuracies of ICL and task vector on bijection tasks (Llama-7B, $n = 10$). We use gray text to indicate accuracies lower than 60%.

Task	Domain $\mathcal{X}$	Domain $\mathcal{Y}$	Example	$\mathcal{X} \rightarrow \mathcal{Y}$		$\mathcal{Y} \rightarrow \mathcal{X}$		$\mathcal{X} \leftrightarrow \mathcal{Y}$	
				ICL	TV	ICL	TV	ICL	TV
To Upper	$\{a, \dots, z\}$	$\{A, \dots, Z\}$	$a \rightarrow A$	1.00	0.91	1.00	0.99	1.00	0.55
Translation	English	French	hello $\rightarrow$ bonjour	0.83	0.84	0.82	0.70	0.54	0.35
	English	Italian	hello $\rightarrow$ ciao	0.84	0.78	0.82	0.74	0.70	0.47
	English	Spanish	hello $\rightarrow$ hola	0.92	0.88	0.89	0.75	0.64	0.43
Linguistic	Present	Gerund	go $\rightarrow$ going	0.99	0.95	1.00	0.97	0.80	0.41
	Present	Past	go $\rightarrow$ went	0.98	0.91	0.99	0.96	0.52	0.33
	Present	Past Perfect	go $\rightarrow$ gone	0.82	0.82	0.94	0.65	0.55	0.33
	Singular	Plural	dog $\rightarrow$ dogs	0.88	0.78	0.94	0.89	0.76	0.51
Copy	$\{a, \dots, z, A, \dots, Z\}$		$A \rightarrow A$	-	-	-	-	1.00	0.98
Antonym	Adjectives		happy $\rightarrow$ sad	0.89	0.83	-	-	0.83	0.73

5 Further Discussions

Inseparable Covariates and Responses. In our main analysis, we assume that x_i and y_i embeddings are linearly separable, allowing the addition $x_i + y_i$ to act a concatenation operation. However, recognizing that this assumption does not generally hold for real-world transformers, we extend our analysis to the following setting, where x_i and y_i are no longer linearly separable. While this still imposes a $2d$ -dimensional requirement on the hidden space, such a constraint is easily satisfied in practical transformers, given the high dimensionality of their internal representations.

$$Z_0 = \begin{bmatrix} 0 & 0 & \dots & 0 & 0 & 0 & 0 \\ x_1 & y_1 & \dots & x_n & y_n & x_{\text{test}} & 0 \end{bmatrix} \in \mathbb{R}^{(2d) \times (2n+2)}. \quad (11)$$

We slightly modify the sparsity constraints for the first layer, and require $(D_0)_{2i,:} = 0$ for $i \in [n+1]$:

$$V_0 = \begin{bmatrix} 0 & A_0 \\ 0_{d \times d} & 0 \end{bmatrix}, \quad Q_0 = \begin{bmatrix} 0_{2d \times 2d} & 0 \\ 0 & D_0 \end{bmatrix}, \quad \text{where } A_0 \in \mathbb{R}^{d \times d}, D_0 \in \mathbb{R}^{d_p \times d_p}. \quad (12)$$

With these conditions, we are ready to establish the critical points for inseparable demonstrations.

Note that V_0 and Q_0 do not involve B_0 and C_0 , so the sequences B and C have size $L-1$.

Theorem 5. Under the same settings as Theorem 1, define $\mathcal{S}_I, \mathcal{S}_\Sigma \subset \mathbb{R}^{d \times d}$ and $\mathcal{S}_P \subset \mathbb{R}^{d_p \times d_p}$ as

$$\mathcal{S}_I = \{\lambda I_d \mid \lambda \in \mathbb{R}\}, \quad \mathcal{S}_\Sigma = \{\lambda \Sigma^{-1} \mid \lambda \in \mathbb{R}\}, \quad \mathcal{S}_P = \{\text{diag}(I_n \otimes \Lambda_1, \Lambda_2) \mid \Lambda_1, \Lambda_2 \in \mathbb{R}^{2 \times 2}\}.$$

Consider optimizing an L -layer linear transformer with inseparable pairwise demonstrations and parameter configuration given in eq. (12) for the first layer and eq. (6) for the remaining layers, then

$$\inf_{A \in \mathcal{S}_I^L, B \in \mathcal{S}_I^{L-1}, C \in \mathcal{S}_\Sigma^{L-1}, D \in \mathcal{S}_P^L} \sum_{H \in A \cup B \cup C \cup D} \|\nabla_H \mathcal{L}(\{V_l, Q_l\}_{l=1}^L)\|_F^2 = 0.$$

This result suggests that for inseparable demonstrations, the first layer performs a functionally similar concatenation operation by "moving" the embedding of each x_i to the corresponding y_i position. This enables the model to reconstruct the single-token structure without linear separability.

Optimal Weights for Causal Task Vectors. While task vectors naturally emerge in linear transformers, their embeddings do not directly help minimize the ICL risk, as evidenced by the identical performance between pairwise and triplet formatted prompts (Figures 4a and 4b). Instead, we show that task vectors contribute to minimizing the training (i.e., LLM pretraining) risk when token-wise dropout is applied, acting as redundancies for in-context demonstrations that may be randomly dropped during training. This redundancy ensures that essential task information is preserved and continues to facilitate accurate prediction despite partial context loss.

Proposition 6. Under the same settings as Proposition 3, consider adding token-wise dropouts O_l :

$$Z_l = Z_{l-1} O_l + \frac{1}{n} \text{Attn}_{V_l, Q_l}(Z_{l-1}) O_l, \quad \text{where } O_l = \text{diag}(o_l^1, \dots, o_l^{d_p}), \quad o_l^i \stackrel{i.i.d.}{\sim} \text{Bern}(p).$$

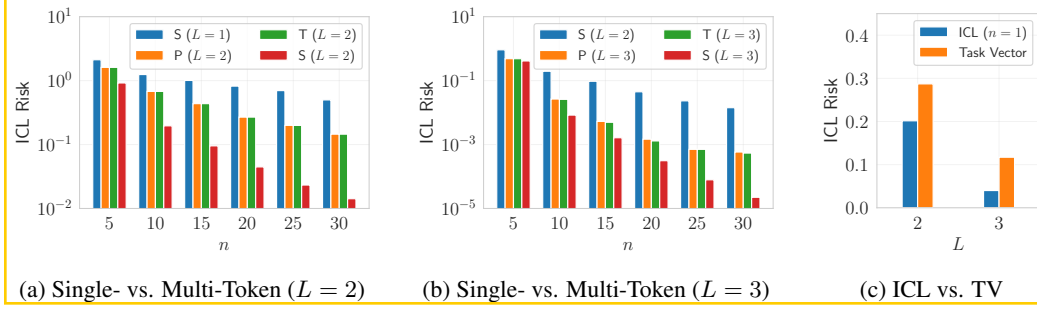


Figure 4: (a, b) Comparison of the best ICL risk achieved using single (S), pairwise (P), and triplet (T) formatted prompts. (c) Performance comparison between 1-shot ICL and task vector.

245 Then any minimizer Λ_4 of the in-context risk $\mathcal{L}(\{V_l, Q_l\}_{l=1}^L)$ satisfies $(\Lambda_4)_{n+1,:} = 0$ and:

$$(\Lambda_4)_{1:n,:} \propto \arg \min_{\Lambda} c_1 \|\Lambda\|_4^4 + c_2 \sum_{i=1}^n \|\Lambda_{i,:}\|_2^4 + c_3 \sum_{j=1}^{n+1} \|\Lambda_{:,j}\|_2^4 + c_4 \|\Lambda \Lambda^\top\|_F^2, \text{ s.t. } \|\Lambda\|_F^2 = 1.$$

246 where $c_1, \dots, c_4$ are non-negative constants depending on V_l, Q_l and p .

247 This result suggests that dropout introduces additional higher-order regularization on the task vec-
 248 tor weights, encouraging them to distribute more uniformly across demonstrations. Furthermore,
 249 when considering causal attention (i.e., enforcing Λ_4 to be upper-triangular), it induces a decaying
 250 weight pattern from later to earlier demonstrations, which is also consistently observed in practical
 251 transformer models as evidenced in Figure 3c. While dropout is not always applied during LLM pre-
 252 training or fine-tuning, the injection of position encodings and use of normalization act as alternative
 253 sources of perturbation, thereby promoting the emergence of such redundancy.

254 **Extra EOS Tokens.** In our theoretical analysis, we consistently impose an additional zero token at
 255 the end of the input prompt. While this token can be interpreted as an EOS token in practical models,
 256 such a design choice is uncommon in standard ICL tasks. We justify this modeling decision with:

257 **Proposition 7** (Informal). *Given any L -layer, single-head, d -dimensional linear-attention trans-*
 258 *former with EOS tokens, there exists an equivalent L -layer, two-head, $2d$ -dimensional linear-attention*
 259 *transformer operating without EOS tokens.*

260 This equivalence suggests that the same learning dynamics can be realized through multi-head
 261 architectures without relying on explicit EOS tokens. Specifically, one head in this setting is dedicated
 262 to task vector formation, while the other handles ICL prediction. This separation allows the model to
 263 retain the functional role of the EOS token implicitly within its hidden states. Consequently, our prior
 264 theoretical analysis can be naturally extended to practical models that omit explicit EOS tokens.

265 6 Experimental Studies

266 6.1 Synthetic Results with Random Linear Regression

267 In this section, we validate our critical points analysis with synthetic linear regression tasks. Specifi-
 268 cally, we examine the achievable ICL risk of linear transformers trained with single-token (eq. (3)),
 269 pairwise (eq. (7)), and triplet (eq. (9)) demonstrations. We set the input dimension to $d = 4$ and
 270 $P_x = P_w = \mathcal{N}(0, I_d)$. For each setting, we train multiple models with different random seeds and
 271 report the minimum ICL risk achieved as a proxy for the global optimum. The comparative results
 272 across different numbers of layers L and demonstration formats are shown in Figures 4a and 4b.

273 These results support our theoretical analysis: when trained with pairwise or triplet demonstrations, the
 274 transformer recovers the GD++ algorithm similar to the single-token case. Notably, the performance
 275 of L -layer transformers with pairwise (P) and triplet (T) demonstrations closely aligns, indicating
 276 a shared underlying learning pattern. Moreover, their performance consistently lies between that
 277 of single-token (S) case L -layer and $(L - 1)$ -layer models. The observed improvement over the
 278 $(L - 1)$ -layer single-token baselines comes from the additional GD++ performed solely on x_n tokens
 279 in the first layer, effectively acting as a "half-step" of gradient descent.

Table 2: Accuracy comparison between standard ICL (Baseline), the task vector method (TaskV), and our strategy (TaskV-M). The experiment is conducted on Llama-13B with $n = 10$.

	Method	Knowledge	Algorithmic	Translation	Linguistic	Bijection	Average
0-shot	Baseline	6.90 $\pm$ 2.08	15.60 $\pm$ 1.72	7.00 $\pm$ 1.65	12.44 $\pm$ 1.74	8.27 $\pm$ 1.33	10.28 $\pm$ 0.98
	TaskV	68.80 $\pm$ 2.66	86.20 $\pm$ 1.61	73.53 $\pm$ 0.91	85.24 $\pm$ 1.80	50.67 $\pm$ 2.32	72.26 $\pm$ 1.01
1-shot	Baseline	69.50 $\pm$ 3.86	73.67 $\pm$ 1.56	57.80 $\pm$ 2.01	56.22 $\pm$ 1.57	44.76 $\pm$ 2.44	58.11 $\pm$ 0.63
	TaskV	79.50 $\pm$ 2.35	88.47 $\pm$ 0.75	80.67 $\pm$ 2.56	89.11 $\pm$ 0.84	60.44 $\pm$ 2.07	78.79 $\pm$ 0.77
	TaskV-M	81.30 $\pm$ 2.80	89.53 $\pm$ 0.65	80.13 $\pm$ 2.14	88.71 $\pm$ 0.62	61.78 $\pm$ 0.96	79.34 $\pm$ 0.37
2-shot	Baseline	78.80 $\pm$ 3.30	85.07 $\pm$ 1.37	75.67 $\pm$ 2.64	76.80 $\pm$ 1.18	56.49 $\pm$ 2.87	72.92 $\pm$ 0.59
	TaskV	84.60 $\pm$ 2.11	88.40 $\pm$ 0.68	84.33 $\pm$ 0.92	90.13 $\pm$ 0.92	62.44 $\pm$ 2.16	80.82 $\pm$ 0.42
	TaskV-M	85.70 $\pm$ 1.63	89.27 $\pm$ 1.10	84.13 $\pm$ 1.15	89.64 $\pm$ 0.86	64.49 $\pm$ 2.02	81.48 $\pm$ 0.37
3-shot	Baseline	86.20 $\pm$ 2.69	88.07 $\pm$ 1.06	80.00 $\pm$ 1.67	84.04 $\pm$ 1.19	62.18 $\pm$ 1.52	78.51 $\pm$ 0.42
	TaskV	90.20 $\pm$ 2.23	88.67 $\pm$ 0.89	86.27 $\pm$ 2.31	92.31 $\pm$ 0.48	66.53 $\pm$ 0.94	83.53 $\pm$ 0.41
	TaskV-M	90.30 $\pm$ 1.50	89.87 $\pm$ 0.83	86.07 $\pm$ 2.17	92.36 $\pm$ 0.72	68.13 $\pm$ 0.76	84.15 $\pm$ 0.52
4-shot	Baseline	84.80 $\pm$ 2.06	88.07 $\pm$ 0.61	83.27 $\pm$ 1.82	88.89 $\pm$ 1.91	67.16 $\pm$ 1.47	81.52 $\pm$ 0.66
	TaskV	88.70 $\pm$ 1.69	89.53 $\pm$ 1.34	86.27 $\pm$ 1.08	92.76 $\pm$ 0.54	70.44 $\pm$ 1.35	84.66 $\pm$ 0.39
	TaskV-M	89.60 $\pm$ 1.43	91.00 $\pm$ 1.01	87.20 $\pm$ 0.62	92.36 $\pm$ 1.44	72.53 $\pm$ 0.94	85.64 $\pm$ 0.29

Additionally, we successfully reproduce the task vector method in linear transformers. Specifically, we extract the hidden state of the final ($\rightarrow$) token from triplet demonstrations after the first layer, and inject this vector into zero-shot prompts consisting of only x_{test} . To simulate the effect of layer normalization used in practical transformers, we normalize the task vectors before inference and the output vectors before ICL risk evaluation. As shown in Figure 4c, the performance of task vectors is parallel to that of standard ICL with a single in-context example. This validates our conjecture that the injected task vector effectively acts as a single demonstration.

6.2 Enhancing the Task Vector Method

We further explore an enhancement to the original task vector method. According to our previous analysis, a single injected task vector may not provide sufficient information for inference on complex tasks (e.g., bijection tasks). Moreover, in linear-attention models, each ($\rightarrow$) token functions as an individual in-context demonstration during the gradient descent phase and thus contributes equally to the ICL risk. Motivated by this, we extend the standard task vector method, which modifies only the final arrow token, and propose a multi-vector variant that injects into every single arrow token in few-shot prompts. This enriched injection scheme enables the model to leverage multiple new demonstrations, thereby providing a more informative and distributed context for prediction.

We compare our multi-vector injection strategy (TaskV-M) against standard N -shot ICL (Baseline) and the original task vector method (TaskV). For each N -shot prompt, we generate $N + 1$ distinct ICL prompts to produce $N + 1$ task vectors, which are then used to replace the embeddings of all arrow tokens in the input. For each task, performance is evaluated over 50 randomly sampled prompts, with mean accuracy and standard deviation reported across 5 independent trials. The final results, summarized in Table 2, span a diverse set of ICL task types, including Knowledge, Algorithmic, Translation, Linguistic, and Bijection, showing that TaskV-M consistently outperforms TaskV, especially on the more challenging bijection tasks. These findings support our analysis that every arrow token contributes meaningfully to the model’s ICL capability.

7 Conclusion

This paper proposes the linear combination conjecture as a plausible explanation for the emergence and functionality of task vectors in ICL. We support this conjecture with both empirical observations and theoretical analysis, demonstrating how task vectors naturally arise under triplet-formatted demonstrations in simple linear transformer models, and why this method inherently fails on general bijection tasks. While the conjecture may not yet offer a complete characterization of ICL dynamics, it provides a new perspective on the underlying mechanisms and offers a promising direction for interpreting intermediate hidden states in modern transformer-based language models.

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395 A Auxiliary Lemmas

396 **Lemma 8** (Proposed in [1]). Given positive objective function $f(A)$ taking parameters $A = \{A_i\}_{i=1}^n$,
 397 where $A_i \in \mathbb{R}^{d_i \times d_i}$. Let $\mathcal{S} = \prod_{i=1}^n \mathcal{S}_i \subset \prod_{i=1}^n \mathbb{R}^{d_i \times d_i}$ be a predefined parameter subspace. Define
 398 $\tilde{A}(t, R_i) = \{A_1, \dots, A_i + tR_i, \dots, A_n\}$ given $i \in [1, n]$, $R_i \in \mathbb{R}^{d_i \times d_i}$ and $t \in \mathbb{R}$. If for any $A \in \mathcal{S}$
 399 and $R_i \in \mathbb{R}^{d_i \times d_i}$, there exists $\tilde{R}_i \in \mathcal{S}_i$ such that

$$\left. \frac{d}{dt} f(\tilde{A}(t, \tilde{R}_i)) \right|_{t=0} \leq \left. \frac{d}{dt} f(\tilde{A}(t, R_i)) \right|_{t=0},$$

400 then we have

$$\inf_{A \in \mathcal{S}} \sum_{i=1}^n \|\nabla_{A_i} f(A)\|_F^2 = 0.$$

401 *Proof.* This lemma is proved as part of the main theorems in [1]. We rearrange the proof here to
 402 accommodate arbitrary function of matrices. Firstly, notice that for any $R = \{R_i\}_{i=1}^n \in \prod_{i=1}^n \mathbb{R}^{d_i \times d_i}$,

$$\sum_{i=1}^n \left. \frac{d}{dt} f(\tilde{A}(t, \tilde{R}_i)) \right|_{t=0} = \left. \frac{d}{dt} f(A + tR) \right|_{t=0}.$$

403 Therefore, the provided precondition is equivalent to stating that for any $A \in \mathcal{S}$ and $R \in \prod_{i=1}^n \mathbb{R}^{d_i \times d_i}$,
 404 there exists $\tilde{R} \in \mathcal{S}$ such that:

$$\left. \frac{d}{dt} f(A + t\tilde{R}) \right|_{t=0} \leq \left. \frac{d}{dt} f(A + tR) \right|_{t=0}.$$

405 Let $R = -\nabla_A f(A)$, we then have

$$\begin{aligned} \left. \frac{d}{dt} f(A + tR) \right|_{t=0} &= \left\langle \frac{df(A - t\nabla_A f(A))}{d(A - t\nabla_A f(A))}, \frac{d(A - t\nabla_A f(A))}{t} \right\rangle \Big|_{t=0} \\ &= \langle \nabla_A f(A), -\nabla_A f(A) \rangle = -\|\nabla_A f(A)\|_F^2. \end{aligned}$$

406 If the infimum of $\|\nabla_A f(A)\|_F^2$ is not zero but some positive value p , then the $\mathcal{S}$ -constrained gradient
 407 flow induced by $\tilde{R}$ will lead to unbounded descent:

$$\left. \frac{d}{dt} f(A + t\tilde{R}) \right|_{t=0} \leq -p.$$

408 This contradicts the fact that $f(A) \geq 0$ and concludes the proof. $\square$

409 The following lemma is an extension of Lemma 5 in [1] by accommodating multivariate y samples as
 410 well as enabling a wider range of demonstration and transformer parameter configurations.

411 **Lemma 9.** Let $x_1, \dots, x_{n+1}$ be i.i.d. samples from an input distribution, and let W be sampled
 412 independently of $\{x_i\}_{i=1}^{n+1}$. Let $Z_0 \in \mathbb{R}^{(2d) \times N}$, where $N \in \mathbb{Z}$ be constructed of form

$$Z_0 = \begin{bmatrix} * & \cdots & * & * \\ * & \cdots & * & 0_d \end{bmatrix} \in \mathbb{R}^{(2d) \times N},$$

413 where the $*$ parts can be arbitrarily constructed from $\{x_i\}_{i=1}^{n+1}$ and W . Let $\tilde{Z}_0$ be defined as replacing
 414 the zero part of Z_0 by y_{n+1} .

$$\tilde{Z}_0 = \begin{bmatrix} * & \cdots & * & * \\ * & \cdots & * & y_{n+1} \end{bmatrix} \in \mathbb{R}^{(2d) \times N}.$$

415 Let $\tilde{Z}_l$ be the output of the l -th layer of the linear transformer, and let $\tilde{X}_l, \tilde{Y}_l \in \mathbb{R}^{d \times N}$ be the first and
 416 last d rows of $\tilde{Z}_l$ respectively. Suppose that the $\{Q_l\}_{l=1}^L$ matrices are of form

$$Q_l = \begin{bmatrix} * & 0_{(2d+d_p) \times d} & * \\ \underbrace{\quad}_{d \text{ columns}} & & \underbrace{\quad}_{d_p \text{ columns}} \end{bmatrix},$$

417 Then the in-context risk of this L -layer linear transformer is equivalent to

$$\mathcal{L}(\{V_l, Q_l\}_{l=1}^L) = \mathbb{E}_{\tilde{Z}_0, W} \left[\text{tr} \left((I_N - M) \tilde{Y}_L^\top \tilde{Y}_L (I_N - M) \right) \right]. \quad (13)$$

418 *Proof.* Let the V_l and Q_l matrices be represented as:

$$V_l = \begin{bmatrix} V_l^1 \\ V_l^2 \end{bmatrix}, \quad Q_l = \begin{bmatrix} Q_l^1 & 0 & Q_l^2 \end{bmatrix},$$

419 where $V_l^1, V_l^2 \in \mathbb{R}^{d \times 2d}$, $Q_l^1 \in \mathbb{R}^{(2d+d_p) \times d}$, $Q_l^2 \in \mathbb{R}^{(2d+d_p) \times d_p}$. Then the update rule in eq. (5) can
420 be rephrased as

$$\begin{aligned} X_l &= X_{l-1} + \frac{1}{n} V_l^1 Z_{l-1} M [Z_{l-1}^\top, P] (Q_l^1 X_{l-1} + Q_l^2 P), \\ Y_l &= Y_{l-1} + \frac{1}{n} V_l^2 Z_{l-1} M [Z_{l-1}^\top, P] (Q_l^1 X_{l-1} + Q_l^2 P). \end{aligned}$$

421 Let $\Delta_Z = \tilde{Z}_0 - Z_0$ i.e. an all-zero matrix except that the last half of the last column is y_{n+1} . Let
422 Δ_X and Δ_Y be its first and last d rows respectively, then $\Delta_X = 0$ and $\Delta_Y = [0 \cdots 0 \ y_{n+1}]$.
423 Note that $\tilde{Z}_l = Z_l + \Delta_Z$ holds for $l = 0$ trivially. Now suppose it holds for some $l = k - 1$, then

$$\begin{aligned} \tilde{X}_k &= \tilde{X}_{k-1} + \frac{1}{n} V_k^1 \tilde{Z}_{k-1} M [\tilde{Z}_{k-1}^\top, P] (Q_k^1 \tilde{X}_{k-1} + Q_k^2 P) \\ &= X_{k-1} + \frac{1}{n} V_k^1 Z_{k-1} M [Z_{k-1}^\top, P] (Q_k^1 X_{k-1} + Q_k^2 P) \\ &\quad + \frac{1}{n} V_k^1 \Delta_Z M [Z_{k-1}^\top, P] (Q_k^1 X_{k-1} + Q_k^2 P) \\ &\quad + \frac{1}{n} V_k^1 Z_{k-1} M [\Delta_Z^\top, 0_{d_p \times d_p}] (Q_k^1 X_{k-1} + Q_k^2 P) \\ &\quad + \frac{1}{n} V_k^1 \Delta_Z M [\Delta_Z^\top, 0_{d_p \times d_p}] (Q_k^1 X_{k-1} + Q_k^2 P) \\ &= X_{k-1} + \frac{1}{n} V_k^1 Z_{k-1} M [Z_{k-1}^\top, P] (Q_k^1 X_{k-1} + Q_k^2 P) = X_k, \end{aligned}$$

424 where the last step holds by noticing that $\Delta_Z M = 0$. Similarly, one can prove that

$$\tilde{Y}_k = Y_{k-1} + \Delta_Y + \frac{1}{n} V_k^2 Z_{k-1} M [Z_{k-1}^\top, P] (Q_k^1 X_{k-1} + Q_k^2 P) = Y_k + \Delta_Y.$$

425 Therefore, it holds that for any $l \in [1, L]$, $\tilde{Z}_l = Z_l + \Delta_Z$. Recall the in-context risk in eq. (2):

$$\begin{aligned} \mathcal{L}(\{V_l, Q_l\}_{l=1}^L) &= \mathbb{E}_{Z_0, W} \|(Z_L)_{(d+1:2d), N} + y_{n+1}\|_2^2 \\ &= \mathbb{E}_{Z_0, W} \|(Y_L + \Delta_Y)(I_N - M)\|_2^2 \\ &= \mathbb{E}_{\tilde{Z}_0, W} \left[\text{tr} \left((I_N - M) \tilde{Y}_L^\top \tilde{Y}_L (I_N - M) \right) \right]. \end{aligned}$$

426 The proof is complete. □

427 B Proof of Theoretical Results

428 B.1 Proof of Proposition 4

429 *Proof.* We will first prove sufficiency. Let $W = ab^\top$ be a rank-one matrix, where $a, b \in \mathbb{R}^d$. The
430 given conditions imply that $x = Wy = WWx = ab^\top ab^\top x$, we then have $b^\top x = b^\top ab^\top ab^\top x =$
431 $(b^\top a)^2 b^\top x$. Since $b^\top x \neq 0$, we can conclude that $b^\top a = \pm 1$. Then, $x = ab^\top ab^\top x = \pm ab^\top x = \pm y$.

432 To prove the necessity, it suffices to show that selecting $W = xx^\top / \|x\|_2^2$ when $x = y$ satisfies the
433 given conditions (alternatively, select $W = -xx^\top / \|x\|_2^2$ when $x = -y$). □

434 B.2 Proof of Theorem 1

435 *Proof.* To enhance the readability of the notations in this proof, we will drop the constant $\frac{1}{n}$ factor
436 in linear attention. Furthermore, we will simplify $\tilde{Z}_0, \tilde{X}_0$ and $\tilde{Y}_0$ in Lemma 9 as Z_0, X_0 and Y_0

respectively. This results in different definitions compared to the original ones, but we will not refer to the original definitions in the remainder of this proof.

$$Z_0 = \begin{bmatrix} X_0 \\ Y_0 \end{bmatrix} = \begin{bmatrix} x_1 & 0 & \cdots & x_n & 0 & x_{\text{test}} & 0 \\ 0 & y_1 & \cdots & 0 & y_n & 0 & y_{\text{test}} \end{bmatrix} \in \mathbb{R}^{(2d) \times (2n+2)}.$$

Let Z_l be the output of the l -th layer of the transformer, and let $X_l, Y_l \in \mathbb{R}^{d \times (2n+2)}$ denote the first and last d rows of Z_l , respectively. Under the constraint in eq. (6), we can verify that

$$\begin{aligned} X_l &= X_{l-1} + A_l X_{l-1} M (X_{l-1}^\top C_l X_{l-1} + D_l), \\ Y_l &= Y_{l-1} + B_l Y_{l-1} M (X_{l-1}^\top C_l X_{l-1} + D_l). \end{aligned} \quad (14)$$

In the following analysis, we will use $f(A \leftarrow B)$ to denote the result of the function f of A when replacing the value of A with B . Additionally, we denote $f(A \leftarrow B * A)$ as $f(A \leftarrow^* B)$ for any operator $*$. Therefore, $f(A \leftarrow^+ B) = f(A \leftarrow A + B)$. We also denote $f(A \leftarrow^{\times} B) = f(A \leftarrow BA)$ and $f(A \leftarrow^{\div} B) = f(A \leftarrow AB)$ for convenience.

Our goal is proving that, for any $E \in A \cup B \cup C \cup D$ and an arbitrary matrix $R \in \mathbb{R}^{d \times d}$ ($\mathbb{R}^{d_p \times d_p}$ for D), there exists $\tilde{R} \in \mathcal{S}_l$ ($\mathcal{S}_\Sigma$ for $C, \mathcal{S}_P$ for D) such that

$$\left. \frac{d}{dt} \mathcal{L}(E \leftarrow^+ t\tilde{R}) \right|_{t=0} \leq \left. \frac{d}{dt} \mathcal{L}(E \leftarrow^+ tR) \right|_{t=0}. \quad (15)$$

Let $\bar{X}_0 = [0, x_1, \dots, 0, x_{\text{test}}]$ be a function of X_0 , we then have $Y_0 = W \bar{X}_0$. Let $U_\perp \in \mathbb{R}^{d \times d}$ be a uniformly sampled random orthonormal matrix, and let $U_\Sigma = \Sigma^{1/2} U_\perp \Sigma^{-1/2}$. One can verify that $U_\Sigma^{-1} = \Sigma^{1/2} U_\perp^\top \Sigma^{-1/2}$. By applying Lemma 9 and the fact that $X_0 \stackrel{d}{=} U_\Sigma \bar{X}_0$, we have that for any given matrix R ,

$$\begin{aligned} & \left. \frac{d}{dt} \mathcal{L}(E \leftarrow^+ tR) \right|_{t=0} \\ &= \left. \frac{d}{dt} \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) Y_L^\top (E \leftarrow^+ tR) Y_L (E \leftarrow^+ tR) (I - M) \right) \right] \right|_{t=0} \\ &= 2 \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) Y_L^\top \left. \frac{d}{dt} Y_L (E \leftarrow^+ tR) \right|_{t=0} (I - M) \right) \right] \\ &= 2 \mathbb{E}_{X_0, W, U_\perp} \left[\text{tr} \left((I - M) Y_L^\top (X_0 \leftarrow^{\times} U_\Sigma) \left. \frac{d}{dt} Y_L (X_0 \leftarrow^{\times} U_\Sigma, E \leftarrow^+ tR) \right|_{t=0} (I - M) \right) \right]. \end{aligned}$$

Next, we will show that eq. (15) holds for each one of A_i, B_i, C_i, D_i for any $i \in [1, L]$.

1. Equation (15) holds for A_i .

We first show that for any $l \in [1, L]$, the following equations hold:

$$X_l (X_0 \leftarrow^{\times} U_\Sigma) = U_\Sigma X_l, \quad (16)$$

$$\left. \frac{d}{dt} X_l (X_0 \leftarrow^{\times} U_\Sigma, A_i \leftarrow^+ tR) \right|_{t=0} = U_\Sigma \left. \frac{d}{dt} X_l (A_i \leftarrow^+ tU_\Sigma^{-1} R U_\Sigma) \right|_{t=0}. \quad (17)$$

It is straightforward to verify that eq. (16) holds for $l = 0$. Now suppose that eq. (16) holds for some $l = k - 1$, we then have

$$\begin{aligned} & X_k (X_0 \leftarrow^{\times} U_\Sigma) \\ &= X_{k-1} (X_0 \leftarrow^{\times} U_\Sigma) + A_l X_{k-1} (X_0 \leftarrow^{\times} U_\Sigma) M \left(X_{k-1}^\top (X_0 \leftarrow^{\times} U_\Sigma) C_l X_{k-1} (X_0 \leftarrow^{\times} U_\Sigma) + D_l \right) \\ &= U_\Sigma X_{k-1} + A_l U_\Sigma X_{k-1} M \left(X_{k-1}^\top U_\Sigma^\top C_l U_\Sigma X_{k-1} + D_l \right) \\ &= U_\Sigma (X_{k-1} + A_l X_{k-1} M (X_{k-1}^\top C_l X_{k-1} + D_l)) = U_\Sigma X_k, \end{aligned}$$

where the third equality follows by noticing that when $A_l = a_l I_d$ and $C_l = c_l \Sigma^{-1}$, we have $A_l U_\Sigma = U_\Sigma A_l$ and $U_\Sigma^\top C_l U_\Sigma = C_l$. This concludes the proof of eq. (16).

458 We now turn to the proof of eq. (17). Notice that when $l < i$, we naturally have

$$\left. \frac{d}{dt} X_l(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} = U_\Sigma \left. \frac{d}{dt} X_l(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) \right|_{t=0} = 0.$$

459 When $l = i$, it is easy to verify that

$$\begin{aligned} \left. \frac{d}{dt} X_l(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} &= RU_\Sigma X_{l-1} M(X_{l-1}^\top U_\Sigma^\top C_l U_\Sigma X_{l-1} + D_l) \\ &= U_\Sigma \cdot U_\Sigma^{-1} RU_\Sigma M(X_{l-1}^\top C_l X_{l-1} + D_l) \\ &= U_\Sigma \left. \frac{d}{dt} X_l(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) \right|_{t=0}. \end{aligned}$$

460 Now suppose that eq. (17) holds for some $l = k - 1 > i$, one can verify that:

$$\begin{aligned} &\left. \frac{d}{dt} X_k(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ &= \left. \frac{d}{dt} X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} + \left. \frac{d}{dt} A_k X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) M \right. \\ &\quad \cdot \left. \left(X_{k-1}^\top(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) C_k X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) + D_k \right) \right|_{t=0} \\ &= \left. \frac{d}{dt} X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ &\quad + A_k \left. \frac{d}{dt} X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} M \left(X_{k-1}^\top(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) C_k X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) + D_k \right) \\ &\quad + A_k X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) M \left. \frac{d}{dt} X_{k-1}^\top(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} C_k X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) \\ &\quad + A_k X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) M X_{k-1}^\top(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) C_k \left. \frac{d}{dt} X_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ &= U_\Sigma \left. \frac{d}{dt} X_{k-1}(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) \right|_{t=0} \\ &\quad + U_\Sigma A_k \left. \frac{d}{dt} X_{k-1}(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) \right|_{t=0} M (X_{k-1}^\top C_k X_{k-1} + D_k) \\ &\quad + U_\Sigma A_k X_{k-1} M \left. \frac{d}{dt} X_{k-1}^\top(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) \right|_{t=0} C_k X_{k-1} \\ &\quad + U_\Sigma A_k X_{k-1} M X_{k-1}^\top C_k \left. \frac{d}{dt} X_{k-1}(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) \right|_{t=0} \\ &= U_\Sigma \left. \frac{d}{dt} X_{k-1}(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) \right|_{t=0} + U_\Sigma \left. \frac{d}{dt} A_k X_{k-1}(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) M \right. \\ &\quad \cdot \left. \left(X_{k-1}^\top(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) C_k X_{k-1}(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) + D_k \right) \right|_{t=0} \\ &= U_\Sigma \left. \frac{d}{dt} X_k(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1}RU_\Sigma) \right|_{t=0}. \end{aligned}$$

461 This completes the proof of eq. (17).

462 Under the condition that $B_l = b_l I_d$ for some $b_l \in \mathbb{R}$, we can simplify eq. (14) as

$$\begin{aligned} Y_l &= Y_{l-1} + b_l Y_{l-1} M(X_{l-1}^\top C_l X_{l-1} + D_l) \\ &= Y_{l-1} (I + b_l M(X_{l-1}^\top C_l X_{l-1} + D_l)) \\ &= Y_0 \prod_{j=1}^l (I + b_j M(X_{j-1}^\top C_j X_{j-1} + D_j)). \end{aligned}$$

463 Define $G_l = \bar{X}_0 \prod_{j=1}^l (I + b_j M(X_{j-1}^\top C_j X_{j-1} + D_j))$, then it satisfies that $Y_l = W G_l$. We are
 464 ready to prove that similar results to eqs. (16) and (17) also hold for $G_l, l \in [1, L]$:

$$G_l(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) = U_\Sigma G_l, \quad (18)$$

$$\left. \frac{d}{dt} G_l(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} = U_\Sigma \left. \frac{d}{dt} G_l(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1} R U_\Sigma) \right|_{t=0}. \quad (19)$$

465 Notice that eq. (18) holds trivially for $l=0$ as $G_0 = \bar{X}_0$. Now suppose that eq. (18) holds for some
 466 $l = k-1$, we then have

$$\begin{aligned} G_k(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) &= G_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) \left(I + b_k M(X_{k-1}^\top (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) C_k X_{k-1} (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) + D_k) \right) \\ &= U_\Sigma G_{k-1} (I + b_k M(X_{k-1}^\top C_k X_{k-1} + D_k)) = U_\Sigma G_k. \end{aligned}$$

467 This concludes eq. (18). As for eq. (19), notice that both sides equal 0 when $l < i$. Now suppose that
 468 eq. (19) holds for some $l = k-1 > i$, we then have:

$$\begin{aligned} & \left. \frac{d}{dt} G_k(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ &= \left. \frac{d}{dt} G_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} + \left. \frac{d}{dt} b_k G_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) M \right. \\ & \quad \cdot \left(X_{k-1}^\top (X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) C_k X_{k-1} (X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) + D_k \right) \Big|_{t=0} \\ &= \left. \frac{d}{dt} G_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ & \quad + b_k \left. \frac{d}{dt} G_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} M \left(X_{k-1}^\top (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) C_k X_{k-1} (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) + D_k \right) \\ & \quad + b_k G_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) M \left. \frac{d}{dt} X_{k-1}^\top (X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} C_k X_{k-1} (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) \\ & \quad + b_k G_{k-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) M X_{k-1}^\top (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) C_k \left. \frac{d}{dt} X_{k-1} (X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ &= U_\Sigma \left. \frac{d}{dt} G_{k-1}(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1} R U_\Sigma) \right|_{t=0} \\ & \quad + b_k U_\Sigma \left. \frac{d}{dt} G_{k-1}(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1} R U_\Sigma) \right|_{t=0} M \left(X_{k-1}^\top C_k X_{k-1} + D_k \right) \\ & \quad + b_k U_\Sigma G_{k-1} M \left. \frac{d}{dt} X_{k-1}^\top (A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1} R U_\Sigma) \right|_{t=0} C_k X_{k-1} \\ & \quad + b_k U_\Sigma G_{k-1} M X_{k-1}^\top C_k \left. \frac{d}{dt} X_{k-1} (A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1} R U_\Sigma) \right|_{t=0} \\ &= U_\Sigma \left. \frac{d}{dt} G_k(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1} R U_\Sigma) \right|_{t=0}. \end{aligned}$$

469 This concludes the proof of eq. (19). Consider the in-context risk:

$$\begin{aligned} & \left. \frac{d}{dt} \mathcal{L}(A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ &= 2 \mathbb{E}_{X_0, W, U_\perp} \left[\text{tr} \left((I - M) Y_L^\top (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) \left. \frac{d}{dt} Y_L(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, A_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} (I - M) \right) \right] \\ &= 2 \mathbb{E}_{X_0, W, U_\perp} \left[\text{tr} \left((I - M) G_L^\top U_\Sigma^\top W^\top W U_\Sigma \left. \frac{d}{dt} G_L(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1} R U_\Sigma) \right|_{t=0} (I - M) \right) \right] \\ &= 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \Sigma^{-1} \left. \frac{d}{dt} \mathbb{E}_{U_\perp} [G_L(A_i \stackrel{\pm}{\leftarrow} tU_\Sigma^{-1} R U_\Sigma)] \right|_{t=0} (I - M) \right) \right] \end{aligned}$$

$$\begin{aligned}
&= 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \Sigma^{-1} \frac{d}{dt} G_L(A_i \leftarrow^\pm \mathbb{E}_{U_\perp} [t U_\Sigma^{-1} R U_\Sigma]) \Big|_{t=0} (I - M) \right) \right] \\
&= 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \Sigma^{-1} \frac{d}{dt} G_L(A_i \leftarrow^\pm \text{tr} I_d) \Big|_{t=0} (I - M) \right) \right] \\
&= \frac{d}{dt} \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) Y_L^\top (A_i \leftarrow^\pm \text{tr} I_d) Y_L(A_i \leftarrow^\pm \text{tr} I_d) (I - M) \right) \Big|_{t=0} \right] \\
&= \frac{d}{dt} \mathcal{L}(A_i \leftarrow^\pm \text{tr} I_d) \Big|_{t=0},
\end{aligned}$$

470 where $r = \mathbb{E}_{U_\perp} [U_\Sigma^{-1} R U_\Sigma] = \frac{1}{d} \text{tr}(\Sigma^{-1/2} R \Sigma^{1/2})$, and we used the fact that $U_\Sigma^\top \Sigma^{-1} U_\Sigma = \Sigma^{-1}$,

471 and $\frac{d}{dt} G_L(A_i \leftarrow^\pm tR) \Big|_{t=0}$ is affine in B_i . This concludes that eq. (15) holds for $A_i, i \in [1, L]$.

472 **2. Equation (15) holds for B_i .**

473 From the recursive expressions in eq. (14), we can conclude that the values of X_l do not depend on

474 B_i . Therefore, we naturally have

$$X_l(B_i \leftarrow^\pm tR) = X_l. \quad (20)$$

475 Next, we would like to show that for any $l \in [1, L]$,

$$\mathbb{E}_W \left[W^\top \frac{d}{dt} Y_l(B_i \leftarrow^\pm tR) \Big|_{t=0} \right] = \Sigma^{-1} \frac{d}{dt} G_l(b_i \leftarrow^\pm t \text{tr}(R)) \Big|_{t=0}. \quad (21)$$

476 When $l < i$, we can easily verify eq. (21) since both sides equal 0. When $l = i$, we can get

$$\begin{aligned}
\mathbb{E}_W \left[W^\top \frac{d}{dt} Y_l(B_i \leftarrow^\pm tR) \Big|_{t=0} \right] &= \mathbb{E}_W [W^\top R Y_{l-1} M(X_{l-1}^\top C_l X_{l-1} + D_l)] \\
&= \mathbb{E}_W [W^\top R W] G_{l-1} M(X_{l-1}^\top C_l X_{l-1} + D_l) \\
&= \text{tr}(R) \Sigma^{-1} G_{l-1} M(X_{l-1}^\top C_l X_{l-1} + D_l) \\
&= \Sigma^{-1} \frac{d}{dt} G_l(b_i \leftarrow^\pm t \text{tr}(R)) \Big|_{t=0}.
\end{aligned}$$

477 Suppose that eq. (21) holds for some $l = k - 1 > i$. One can then verify

$$\begin{aligned}
&\mathbb{E}_W \left[W^\top \frac{d}{dt} Y_k(B_i \leftarrow^\pm tR) \Big|_{t=0} \right] \\
&= \mathbb{E}_W \left[W^\top \frac{d}{dt} Y_{k-1}(B_i \leftarrow^\pm tR) (I + b_k M(X_{k-1}^\top C_k X_{k-1} + D_k)) \Big|_{t=0} \right] \\
&= \mathbb{E}_W \left[W^\top \frac{d}{dt} Y_{k-1}(B_i \leftarrow^\pm tR) \Big|_{t=0} \right] (I + b_k M(X_{k-1}^\top C_k X_{k-1} + D_k)) \\
&= \Sigma^{-1} \frac{d}{dt} G_{k-1}(b_i \leftarrow^\pm t \text{tr}(R)) \Big|_{t=0} (I + b_k M(X_{k-1}^\top C_k X_{k-1} + D_k)) \\
&= \Sigma^{-1} \frac{d}{dt} G_k(b_i \leftarrow^\pm t \text{tr}(R)) \Big|_{t=0}.
\end{aligned}$$

478 The proof of eq. (21) is complete. Now, look at the in-context risk, we have

$$\begin{aligned}
\frac{d}{dt} \mathcal{L}(B_i \leftarrow^\pm tR) \Big|_{t=0} &= 2 \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) Y_L^\top \frac{d}{dt} Y_L(B_i \leftarrow^\pm tR) \Big|_{t=0} (I - M) \right) \right] \\
&= 2 \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \mathbb{E}_W \left[W^\top \frac{d}{dt} Y_L(B_i \leftarrow^\pm tR) \Big|_{t=0} \right] (I - M) \right) \right] \\
&= 2 \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \Sigma^{-1} \frac{d}{dt} G_L(b_i \leftarrow^\pm t \text{tr}(R)) \Big|_{t=0} (I - M) \right) \right] \\
&= 2 \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) Y_L^\top \frac{d}{dt} Y_L(B_i \leftarrow^\pm t \text{tr}(R) I_d) \Big|_{t=0} (I - M) \right) \right]
\end{aligned}$$

$$= \frac{d}{dt} \mathcal{L}(B_i \stackrel{\pm}{\leftarrow} t \operatorname{tr}(R) I_d) \Big|_{t=0}.$$

479 This concludes that eq. (15) holds for B_i , $i \in [1, L]$.

480 **3. Equation (15) holds for C_i .**

481 Similar to the A_i case, we will first prove that for any $l \in [1, L]$,

$$\frac{d}{dt} X_l(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, C_i \stackrel{\pm}{\leftarrow} tR) \Big|_{t=0} = U_\Sigma \frac{d}{dt} X_l(C_i \stackrel{\pm}{\leftarrow} tU_\Sigma^\top R U_\Sigma) \Big|_{t=0}. \quad (22)$$

482 The equation above holds trivially for $l < i$. For the case $l = i$, we have

$$\begin{aligned} & \frac{d}{dt} X_l(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, C_i \stackrel{\pm}{\leftarrow} tR) \Big|_{t=0} \\ &= A_j X_{l-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) M X_{l-1}^\top (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) R X_{l-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) \\ &= U_\Sigma A_j X_{l-1} M X_{l-1}^\top U_\Sigma^\top R U_\Sigma X_{l-1} = U_\Sigma \frac{d}{dt} X_l(C_i \stackrel{\pm}{\leftarrow} tU_\Sigma^\top R U_\Sigma) \Big|_{t=0}. \end{aligned}$$

483 One can conclude the proof of eq. (22) through a similar reduction as eq. (17) for $l > i$ layers. Next,
484 we establish the corresponding result for G_i :

$$\frac{d}{dt} G_l(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, C_i \stackrel{\pm}{\leftarrow} tR) \Big|_{t=0} = U_\Sigma \frac{d}{dt} G_l(C_i \stackrel{\pm}{\leftarrow} tU_\Sigma^\top R U_\Sigma) \Big|_{t=0}. \quad (23)$$

485 This equation holds trivially for $l < i$. When taking $l = i$, we can verify that

$$\begin{aligned} & \frac{d}{dt} G_l(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, C_i \stackrel{\pm}{\leftarrow} tR) \Big|_{t=0} = b_l G_{l-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) M X_{l-1}^\top (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) R X_{l-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) \\ &= b_l U_\Sigma G_{l-1}(X_0 \stackrel{\times}{\leftarrow} U_\Sigma) M X_{l-1}^\top U_\Sigma^\top R U_\Sigma X_{l-1} \\ &= U_\Sigma \frac{d}{dt} G_l(C_i \stackrel{\pm}{\leftarrow} tU_\Sigma^\top R U_\Sigma) \Big|_{t=0}. \end{aligned}$$

486 For $l > i$ layers, one can follow similar reductions as eq. (19) to finish the proof. We then consider
487 the in-context risk:

$$\begin{aligned} & \frac{d}{dt} \mathcal{L}(C_i \stackrel{\pm}{\leftarrow} tR) \Big|_{t=0} \\ &= 2 \mathbb{E}_{X_0, W, U_\perp} \left[\operatorname{tr} \left((I - M) Y_L^\top (X_0 \stackrel{\times}{\leftarrow} U_\Sigma) \frac{d}{dt} Y_L(X_0 \stackrel{\times}{\leftarrow} U_\Sigma, C_i \stackrel{\pm}{\leftarrow} tR) \Big|_{t=0} (I - M) \right) \right] \\ &= 2 \mathbb{E}_{X_0, W, U_\perp} \left[\operatorname{tr} \left((I - M) G_L^\top U_\Sigma^\top W^\top W U_\Sigma \frac{d}{dt} G_L(C_i \stackrel{\pm}{\leftarrow} tR) \Big|_{t=0} (I - M) \right) \right] \\ &= 2d \mathbb{E}_{X_0} \left[\operatorname{tr} \left((I - M) G_L^\top \Sigma^{-1} \frac{d}{dt} \mathbb{E}_{U_\perp} [G_L(C_i \stackrel{\pm}{\leftarrow} tU_\Sigma^\top R U_\Sigma)] \Big|_{t=0} (I - M) \right) \right] \\ &= 2d \mathbb{E}_{X_0} \left[\operatorname{tr} \left((I - M) G_L^\top \Sigma^{-1} \frac{d}{dt} G_L(C_i \stackrel{\pm}{\leftarrow} tR \Sigma^{-1}) \Big|_{t=0} (I - M) \right) \right] \\ &= \frac{d}{dt} \mathbb{E}_{X_0, W} \left[\operatorname{tr} \left((I - M) Y_L^\top (C_i \stackrel{\pm}{\leftarrow} tR \Sigma^{-1}) Y_L(C_i \stackrel{\pm}{\leftarrow} tR \Sigma^{-1}) (I - M) \right) \Big|_{t=0} \right] \\ &= \frac{d}{dt} \mathcal{L}(C_i \stackrel{\pm}{\leftarrow} tR \Sigma^{-1}) \Big|_{t=0}, \end{aligned}$$

488 where $r = \mathbb{E}_{U_\perp} [U_\Sigma^\top R U_\Sigma] = \frac{1}{d} \operatorname{tr}(\Sigma^{1/2} R \Sigma^{1/2})$. This concludes that eq. (15) holds for C_i .

489 **4. Equation (15) holds for D_i .**

490 Let $U_p \in \mathbb{R}^{n \times n}$ be a uniformly sampled permutation matrix, i.e., a binary matrix that has exactly one 1
491 entry in each row and column with all other entries 0. Let $U_\circ = \operatorname{diag}(U_p \otimes I_2, I_2) \in \mathbb{R}^{(2n+2) \times (2n+2)}$.

One can verify that by multiplying $X_0 U_\pm$, it is equal to shuffling the first n 2-column sub-blocks of X_0 and keeping the last 2 columns unchanged.

Then, consider a matrix $U_\xi = \text{diag}(\xi_1, \dots, \xi_{n+1}) \in \mathbb{R}^{(n+1) \times (n+1)}$ where $\xi_i \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{\pm 1\}$, i.e., a diagonal matrix with random ± 1 entries. Let $U_\pm = U_\xi \otimes I_2 \in \mathbb{R}^{(2n+2) \times (2n+2)}$. Thus, $U_\pm = U_\pm^\top$ and $X_0 U_\pm$ is randomly flipping the sign of each 2-column sub-block in X_0 .

We are going to prove that for any $l \in [1, L]$, recalling that $f(A \stackrel{\circ}{\leftarrow} B) = f(A \leftarrow AB)$,

$$X_l(X_0 \stackrel{\circ}{\leftarrow} U_\pm U_o) = X_l U_\pm U_o, \quad (24)$$

$$G_l(X_0 \stackrel{\circ}{\leftarrow} U_\pm U_o) = G_l U_\pm U_o. \quad (25)$$

Equation (24) holds trivially for $l = 0$. When eq. (24) holds for some $l = k - 1$, we can verify that

$$\begin{aligned} & X_k(X_0 \stackrel{\circ}{\leftarrow} U_\pm U_o) \\ &= X_{k-1} U_\pm U_o + A_k X_{k-1} U_\pm U_o M (U_o^\top U_\pm^\top X_{k-1}^\top C_k X_{k-1} U_\pm U_o + D_k) \\ &= X_{k-1} U_\pm U_o + A_k X_{k-1} U_\pm U_o M U_o^\top U_\pm^\top (X_{k-1}^\top C_k X_{k-1} + U_\pm U_o D_k U_o^\top U_\pm^\top) U_\pm U_o \\ &= X_{k-1} U_\pm U_o + A_k X_{k-1} M (X_{k-1}^\top C_k X_{k-1} + D_k) U_\pm U_o \\ &= (X_{k-1} + A_k X_{k-1} M (X_{k-1}^\top C_k X_{k-1} + D_k)) U_\pm U_o = X_k U_\pm U_o. \end{aligned}$$

It uses the fact that there exists some $D_i^1, D_i^2 \in \mathbb{R}^{2 \times 2}$ such that $D_i = \text{diag}(I_n \otimes D_i^1, D_i^2)$, so shuffling the first $n \times 2$ diagonal sub-blocks of D_i does not change the matrix, and we have $U_o D_i U_o^\top = D_i$. Similarly, we have $U_\pm D_k U_\pm^\top = D_k$. This concludes eq. (24), and eq. (25) could be acquired similarly.

Next, we will establish the following equalities for X_l and G_l :

$$\left. \frac{d}{dt} X_l(X_0 \stackrel{\circ}{\leftarrow} U_\pm U_o, D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} = \left. \frac{d}{dt} X_l(D_i \stackrel{\pm}{\leftarrow} tU_\pm U_o R U_o^\top U_\pm^\top) \right|_{t=0} U_\pm U_o, \quad (26)$$

$$\left. \frac{d}{dt} G_l(X_0 \stackrel{\circ}{\leftarrow} U_\pm U_o, D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} = \left. \frac{d}{dt} G_l(D_i \stackrel{\pm}{\leftarrow} tU_\pm U_o R U_o^\top U_\pm^\top) \right|_{t=0} U_\pm U_o. \quad (27)$$

The proof follows by similar reductions as proving eqs. (17) and (19).

Finally, we consider the in-context risk under the permutation of U_p and U_ξ . Since each pair of (x_i, y_i) is equivalently sampled from Gaussian distributions, we have $X_0 \stackrel{d}{=} X_0 U_\pm U_o$. Therefore,

$$\begin{aligned} & \left. \frac{d}{dt} \mathcal{L}(D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ &= 2 \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) Y_L^\top \left. \frac{d}{dt} Y_L(D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} (I - M) \right) \right] \\ &= 2 \mathbb{E}_{X_0, W, U_p, U_\xi} \left[\text{tr} \left((I - M) Y_L^\top (X_0 \stackrel{\circ}{\leftarrow} U_\pm U_o) \left. \frac{d}{dt} Y_L(X_0 \stackrel{\circ}{\leftarrow} U_\pm U_o, D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} (I - M) \right) \right] \\ &= 2d \mathbb{E}_{X_0, U_p, U_\xi} \left[\text{tr} \left((I - M) U_o^\top U_\pm^\top G_L^\top \Sigma^{-1} \left. \frac{d}{dt} G_L(D_i \stackrel{\pm}{\leftarrow} tU_\pm U_o R U_o^\top U_\pm^\top) \right|_{t=0} U_\pm U_o (I - M) \right) \right] \\ &= 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \Sigma^{-1} \left. \frac{d}{dt} \mathbb{E}_{U_p, U_\xi} [G_L(D_i \stackrel{\pm}{\leftarrow} tU_\pm U_o^\top R U_o U_\pm)] \right|_{t=0} (I - M) \right) \right] \\ &= 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \Sigma^{-1} \left. \frac{d}{dt} G_L(D_i \stackrel{\pm}{\leftarrow} t\tilde{R}) \right|_{t=0} (I - M) \right) \right] = \left. \frac{d}{dt} \mathcal{L}(D_i \stackrel{\pm}{\leftarrow} t\tilde{R}) \right|_{t=0}, \end{aligned}$$

where $\tilde{R} = \mathbb{E}_{U_p, U_\xi} [U_\pm U_o^\top R U_o U_\pm] = \text{diag}(I_n \otimes R^1, R^2)$, $R^1 = \frac{1}{n} \sum_{j=1}^n R_j$, $R^2 = R_{n+1}$, and R_j is the j -th 2×2 diagonal block of R . The 4th equality uses the fact that $\text{tr}[(I - M)A(I - M)]$ is extracting the right-bottom element of A , so it should be equal to $\text{tr}[(I - M)U_o^\top U_\pm^\top A U_\pm U_o(I - M)]$ for any matrix A . This concludes that eq. (15) holds for D_i .

Till now, we have proved that eq. (15) holds for each one of A_i, B_i, C_i, D_i . The proof of the whole theorem is then completed by applying Lemma 8. $\square$

513 B.3 Proof of Theorem 2

514 *Proof.* In this proof, we follow the same notations as the proof of Theorem 1, where the constant $\frac{1}{n}$
 515 factor is dropped and $\tilde{Z}_0, \tilde{X}_0, \tilde{Y}_0$ are simplified as Z_0, X_0, Y_0 respectively.

$$Z_0 = \begin{bmatrix} x_1 & 0 & 0 & \cdots & x_n & 0 & 0 & x_{\text{test}} & 0 & 0 \\ 0 & 0 & y_1 & \cdots & 0 & 0 & y_n & 0 & 0 & y_{\text{test}} \end{bmatrix} \in \mathbb{R}^{(2d) \times (3n+3)}. \quad (28)$$

516 Let $Z_l \in \mathbb{R}^{2d \times (3n+3)}$ be the l -th layer's output and let $X_l, Y_l \in \mathbb{R}^{d \times (3n+3)}$ be its first and last d
 517 rows. Our goal is to prove that, for any $E \in A \cup B \cup C \cup D$ and an arbitrary matrix $R \in \mathbb{R}^{d \times d}$
 518 ($\mathbb{R}^{d_p \times d_p}$ for D), there exists $\tilde{R} \in \mathcal{S}_l$ ($\mathcal{S}_\Sigma$ for C , $\mathcal{S}_P$ for D) such that

$$\left| \frac{d}{dt} \mathcal{L}(E \stackrel{\pm}{\leftarrow} t\tilde{R}) \right|_{t=0} \leq \left| \frac{d}{dt} \mathcal{L}(E \stackrel{\pm}{\leftarrow} tR) \right|_{t=0}. \quad (29)$$

519 The proofs of eq. (29) for A_i, B_i and C_i are identical with the proof of Theorem 1 so we omit them.
 520 We will be focusing on D_i for the rest of the proof.

521 Let $U_p^s \in \mathbb{R}^{n \times n}$ and $U_p^t \in \mathbb{R}^{(n+1) \times (n+1)}$ be uniformly sampled permutation matrices. Let $U_\circ^s =$
 522 $\text{diag}(U_p^s, 1) \otimes \text{diag}(1, 0, 1)$ and $U_\circ^t = U_p^t \otimes \text{diag}(0, 1, 0)$. Therefore, $X_0 U_\circ^s$ is shuffling the 1-st
 523 and 3-rd columns among each 3-column sub-block of X_0 (except for the last 3-column sub-block),
 524 and $X_0 U_\circ^t$ is shuffling the 2-nd column among each 3-column sub-block. Next, let $U_\xi^s, U_\xi^t \in$
 525 $\mathbb{R}^{(n+1) \times (n+1)}$ be diagonal matrices with uniformly sampled ± 1 entries. Define $U_\pm^s = U_\xi^s \otimes$
 526 $\text{diag}(1, 0, 1)$ and $U_\pm^t = U_\xi^t \otimes \text{diag}(0, 1, 0)$. It can then be verified that $X_0 U_\pm^s U_\pm^t \stackrel{d}{=} X_0$.
 527 To simplify the notations, let $U_\equiv$ denote $U_\pm^s U_\pm^t U_\circ^s U_\circ^t$. We will focus on a subset of $\mathcal{S}_P$:

$$\mathcal{S}'_P = \left\{ \text{diag}(I_n \otimes \Lambda_1, \Lambda_2) + I_{n+1} \otimes \Lambda_3 \mid \Lambda_1, \Lambda_2 \in \mathcal{M} \left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} \right), \Lambda_3 \in \mathcal{M} \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right) \right\}.$$

528 Assume $D_k = \text{diag}(I_n \otimes \Lambda_1, \Lambda_2) + I_{n+1} \otimes \Lambda_3 \in \mathcal{S}'_P$ as defined above, one can verify that it is
 529 a block-diagonal matrix constructed from the same 3×3 sub-blocks, and thus is invariant under
 530 $U_\equiv D_k U_\equiv^\top$. We will then prove that for any $l \in [1, L]$,

$$X_l(X_0 \stackrel{\circ}{\leftarrow} U_\equiv) = X_l U_\equiv, \quad (30)$$

$$G_l(X_0 \stackrel{\circ}{\leftarrow} U_\equiv) = G_l U_\equiv, \quad (31)$$

$$\left| \frac{d}{dt} X_l(X_0 \stackrel{\circ}{\leftarrow} U_\equiv, D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} = \left| \frac{d}{dt} X_l(D_i \stackrel{\pm}{\leftarrow} tU_\equiv R U_\equiv^\top) \right|_{t=0} U_\equiv, \quad (32)$$

$$\left| \frac{d}{dt} G_l(X_0 \stackrel{\circ}{\leftarrow} U_\equiv, D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} = \left| \frac{d}{dt} G_l(D_i \stackrel{\pm}{\leftarrow} tU_\equiv R U_\equiv^\top) \right|_{t=0} U_\equiv. \quad (33)$$

531 These results can be acquired by similar proofs as eqs. (24) to (27). We then consider the in-context
 532 risk under the permutations of $U_\equiv$. Similarly, we have $X_0 \stackrel{d}{=} X_0 U_\equiv$ and

$$\begin{aligned} & \left| \frac{d}{dt} \mathcal{L}(D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ &= 2 \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) Y_L^\top \left| \frac{d}{dt} Y_L(D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} (I - M) \right) \right] \\ &= 2d \mathbb{E}_{X_0, U_\equiv} \left[\text{tr} \left((I - M) G_L^\top(X_0 \stackrel{\circ}{\leftarrow} U_\equiv) \Sigma^{-1} \left| \frac{d}{dt} G_L(X_0 \stackrel{\circ}{\leftarrow} U_\equiv, D_i \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} (I - M) \right) \right] \\ &= 2d \mathbb{E}_{X_0, U_\equiv} \left[\text{tr} \left((I - M) U_\equiv^\top G_L^\top \Sigma^{-1} \left| \frac{d}{dt} G_L(D_i \stackrel{\pm}{\leftarrow} tU_\equiv R U_\equiv^\top) \right|_{t=0} U_\equiv (I - M) \right) \right] \\ &= 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \Sigma^{-1} \left| \frac{d}{dt} G_L(D_i \stackrel{\pm}{\leftarrow} tU_\equiv [U_\equiv R U_\equiv^\top]) \right|_{t=0} (I - M) \right) \right] \\ &= \left| \frac{d}{dt} \mathcal{L}(D_i \stackrel{\pm}{\leftarrow} t\tilde{R}) \right|_{t=0}. \end{aligned}$$

533 Let R_j be the j -th 3×3 diagonal block of R , then $R^1 = \frac{1}{n} \sum_{j=1}^n R_j \circ \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, $R^2 = R_{n+1} \circ \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$
 534 $R^3 = \frac{1}{n+1} \sum_{j=1}^{n+1} R_j \circ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $\tilde{R} = \mathbb{E}_{U \equiv} [U \equiv R U^\top] = \text{diag}(I_n \otimes R^1, R^2) + I_{n+1} \otimes R^3$. This
 535 indicates that eq. (29) holds for each $D_i \in \mathcal{S}'_P$, and thus the proof of the whole theorem completes by
 536 applying Lemma 8 and noticing that $\mathcal{S}'_P \subset \mathcal{S}_P$. $\square$

537 B.4 Proof of Theorem 5

538 *Proof.* We keep the same notations as the proof of Theorem 1, dropping the $\frac{1}{n}$ factor and simplifying
 539 X_0, Y_0, Z_0 as X_0, Y_0, Z_0 , as follows:

$$Z_0 = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ x_1 & y_1 & \cdots & x_n & y_n & x_{\text{test}} & y_{\text{test}} \end{bmatrix} \in \mathbb{R}^{(2d) \times (2n+2)}. \quad (34)$$

540 Note that we now have X_0 and Y_0 containing both x_j and y_j . Define

$$\begin{aligned} X &= [x_1 & 0 & \cdots & x_n & 0 & x_{\text{test}} & 0], \\ \bar{X} &= [0 & x_1 & \cdots & 0 & x_n & 0 & x_{\text{test}}], \\ Y &= [0 & y_1 & \cdots & 0 & y_n & 0 & y_{\text{test}}]. \end{aligned}$$

541 we then have $Y_0 = X + Y = X + W\bar{X}$. From the parameter configuration in eq. (12), the update
 542 rule of the first attention layer is

$$X_1 = A_1 Y_0 M D_1 = A_1 X M D_1, \quad Y_1 = Y_0 = X + W\bar{X}. \quad (35)$$

543 The update rule for the following layers is the same as eq. (14). We are going to prove that, for any
 544 $E \in A \cup B \cup C \cup D$ and an arbitrary matrix $R \in \mathbb{R}^{d \times d}$ ($\mathbb{R}^{d_p \times d_r}$ for D), there exists $\tilde{R} \in \mathcal{S}_E$ ($\mathcal{S}_\Sigma$
 545 for $C, \mathcal{S}_P$ for D) such that

$$\left. \frac{d}{dt} \mathcal{L}(E \stackrel{\pm}{\leftarrow} t\tilde{R}) \right|_{t=0} \leq \left. \frac{d}{dt} \mathcal{L}(E \stackrel{\pm}{\leftarrow} tR) \right|_{t=0}. \quad (36)$$

546 Similarly to Theorem 1, we uniformly sample $U_\perp \in \mathbb{R}^{d \times d}$ as an orthonormal random matrix, and let
 547 $\bar{U}_\Sigma = \Sigma^{1/2} U_\perp \Sigma^{-1/2}$. Under the condition that $B_l = b_l I_d$ for some $b_l \in \mathbb{R}$, we have

$$Y_l = Y_1 \prod_{j=2}^l (I + b_j M (X_{j-1}^\top C_j X_{j-1} + D_j)).$$

548 Let $F_l = X \prod_{j=2}^l (I + b_j M (X_{j-1}^\top C_j X_{j-1} + D_j))$, $G_l = \bar{X} \prod_{j=2}^l (I + b_j M (X_{j-1}^\top C_j X_{j-1} + D_j))$,
 549 we then have $Y_l = F_l + W G_l$. According to Lemma 9,

$$\begin{aligned} & \left. \frac{d}{dt} \mathcal{L}(E \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} \\ &= \left. \frac{d}{dt} \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) Y_L^\top (E \stackrel{\pm}{\leftarrow} tR) Y_L (E \stackrel{\pm}{\leftarrow} tR) (I - M) \right) \right] \right|_{t=0} \\ &= \left. \frac{d}{dt} \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) F_L^\top (E \stackrel{\pm}{\leftarrow} tR) F_L (E \stackrel{\pm}{\leftarrow} tR) (I - M) \right) \right] \right|_{t=0} \\ &\quad + \left. \frac{d}{dt} \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) G_L^\top (E \stackrel{\pm}{\leftarrow} tR) W^\top W G_L (E \stackrel{\pm}{\leftarrow} tR) (I - M) \right) \right] \right|_{t=0} \\ &= 2 \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) F_L^\top \left. \frac{d}{dt} F_L (E \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} (I - M) \right) \right] \\ &\quad + 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \Sigma^{-1} \left. \frac{d}{dt} G_L (E \stackrel{\pm}{\leftarrow} tR) \right|_{t=0} (I - M) \right) \right]. \end{aligned}$$

550 Next, we will show that eq. (36) holds for each one of A_i, B_i, C_i, D_i for any $i \in [1, L]$.

551 **1. Equation (36) holds for A_i .**

552 One can easily verify that eqs. (16) and (17) still hold. Furthermore, eqs. (18) and (19) hold for both
 553 F and G . With these observations, we can then verify

$$\begin{aligned}
 & \left. \frac{d}{dt} \mathcal{L}(A_i \leftarrow^\pm tR) \right|_{t=0} \\
 &= 2 \mathbb{E}_{X_0, U_\perp} \left[\text{tr} \left((I - M) F_L^\top (X \leftarrow U_\Sigma) \frac{d}{dt} F_L(X \leftarrow U_\Sigma, A_i \leftarrow^\pm tR) \Big|_{t=0} (I - M) \right) \right] \\
 &\quad + 2d \mathbb{E}_{X_0, U_\perp} \left[\text{tr} \left((I - M) G_L^\top (X \leftarrow U_\Sigma) \Sigma^{-1} \frac{d}{dt} G_L(X \leftarrow U_\Sigma, A_i \leftarrow^\pm tR) \Big|_{t=0} (I - M) \right) \right] \\
 &= 2 \mathbb{E}_{X_0, U_\perp} \left[\text{tr} \left((I - M) F_L^\top U_\Sigma^\top U_\Sigma \frac{d}{dt} F_L(A_i \leftarrow^\pm tU_\Sigma^{-1} R U_\Sigma) \Big|_{t=0} (I - M) \right) \right] \\
 &\quad + 2d \mathbb{E}_{X_0, U_\perp} \left[\text{tr} \left((I - M) G_L^\top U_\Sigma^\top \Sigma^{-1} U_\Sigma \frac{d}{dt} G_L(A_i \leftarrow^\pm tU_\Sigma^{-1} R U_\Sigma) \Big|_{t=0} (I - M) \right) \right] \\
 &= 2 \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) F_L^\top \frac{d}{dt} F_L(A_i \leftarrow^\pm tR I_d) \Big|_{t=0} (I - M) \right) \right] \\
 &\quad + 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_L^\top \Sigma^{-1} \frac{d}{dt} G_L(A_i \leftarrow^\pm tR I_d) \Big|_{t=0} (I - M) \right) \right] \\
 &= \left. \frac{d}{dt} \mathcal{L}(A_i \leftarrow^\pm tR I_d) \right|_{t=0},
 \end{aligned}$$

554 where $r = \mathbb{E}_{U_\perp} [U_\Sigma^{-1} R U_\Sigma] = \frac{1}{d} \text{tr}(\Sigma^{-1/2} R \Sigma^{1/2})$.

555 **2. Equation (36) holds for B_i .**

556 From the definition of F and G , we can verify that

$$\begin{aligned}
 & \left. \frac{d}{dt} Y_l(B_i \leftarrow^\pm tR) \right|_{t=0} \\
 &= R(F_{i-1} + W G_{i-1}) M(X_{i-1}^\top C_i X_{i-1} + D_i) \prod_{j=i+1}^l (I + b_j M(X_{j-1}^\top C_j X_{j-1} + D_j)).
 \end{aligned}$$

557 Define

$$\begin{aligned}
 \bar{F}_l^i &= (F_{i-1} + B_i F_{i-1} M(X_{i-1}^\top C_i X_{i-1} + D_i)) \prod_{j=i+1}^l (I + b_j M(X_{j-1}^\top C_j X_{j-1} + D_j)), \\
 \bar{G}_l^i &= (W G_{i-1} + B_i W G_{i-1} M(X_{i-1}^\top C_i X_{i-1} + D_i)) \prod_{j=i+1}^l (I + b_j M(X_{j-1}^\top C_j X_{j-1} + D_j)),
 \end{aligned}$$

558 We then have

$$\left. \frac{d}{dt} Y_l(B_i \leftarrow^\pm tR) \right|_{t=0} = \left. \frac{d}{dt} \bar{F}_l^i(B_i \leftarrow^\pm tR) \right|_{t=0} + \left. \frac{d}{dt} \bar{G}_l^i(B_i \leftarrow^\pm tR) \right|_{t=0}.$$

559 Similar to eqs. (19) and (21), we can prove that

$$\begin{aligned}
 & \left. \frac{d}{dt} \bar{F}_l^i(X_0 \leftarrow U_\Sigma, B_i \leftarrow^\pm tR) \right|_{t=0} = U_\Sigma \left. \frac{d}{dt} \bar{F}_l^i(B_i \leftarrow^\pm tU_\Sigma^{-1} R U_\Sigma) \right|_{t=0}, \\
 & \mathbb{E}_W \left[W^\top \left. \frac{d}{dt} \bar{G}_l^i(B_i \leftarrow^\pm tR) \right|_{t=0} \right] = \Sigma^{-1} \left. \frac{d}{dt} \bar{G}_l^i(B_i \leftarrow^\pm t \text{tr}(R) I_d) \right|_{t=0}.
 \end{aligned}$$

560 Without loss of generality, we assume that $r = \frac{1}{d} \text{tr}(\Sigma^{-1/2} R \Sigma^{1/2}) \leq \frac{1}{d} \text{tr}(R)$, and let $\gamma =$

561 $r d / \text{tr}(R) \leq 1$. Then, one can verify that

$$\left. \frac{d}{dt} \mathcal{L}(B_i \leftarrow^\pm tR) \right|_{t=0}$$

$$\begin{aligned}
&= 2 \mathbb{E}_{X_0, U_\perp} \left[\text{tr} \left((I - M) F_l^\top (X \stackrel{\times}{\leftarrow} U_\Sigma) \frac{d}{dt} \bar{F}_l^i (X \stackrel{\times}{\leftarrow} U_\Sigma, B_i \stackrel{\pm}{\leftarrow} tR) \Big|_{t=0} (I - M) \right) \right] \\
&\quad + 2 \mathbb{E}_{X_0, W} \left[\text{tr} \left((I - M) G_l^\top W^\top \frac{d}{dt} \bar{G}_l^i (B_i \stackrel{\pm}{\leftarrow} tR) \Big|_{t=0} (I - M) \right) \right] \\
&= 2 \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) F_l^\top \frac{d}{dt} \bar{F}_l^i (B_i \stackrel{\pm}{\leftarrow} \text{tr} I_d) \Big|_{t=0} (I - M) \right) \right] \\
&\quad + 2 \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_l^\top \Sigma^{-1} \frac{d}{dt} \bar{G}_l^i (B_i \stackrel{\pm}{\leftarrow} t \text{tr}(R) I_d) \Big|_{t=0} (I - M) \right) \right] \\
&= 2 \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) F_l^\top \frac{d}{dt} F_l (B_i \stackrel{\pm}{\leftarrow} \text{tr} I_d) \Big|_{t=0} (I - M) \right) \right] \\
&\quad + \frac{1}{\gamma} 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_l^\top \Sigma^{-1} \frac{d}{dt} G_l (B_i \stackrel{\pm}{\leftarrow} \text{tr} I_d) \Big|_{t=0} (I - M) \right) \right] \\
&= \left(\frac{1}{\gamma} - 1 \right) 2d \mathbb{E}_{X_0} \left[\text{tr} \left((I - M) G_l^\top \Sigma^{-1} \frac{d}{dt} G_l (B_i \stackrel{\pm}{\leftarrow} \text{tr} I_d) \Big|_{t=0} (I - M) \right) \right] \\
&\quad + \frac{d}{dt} \mathcal{L}(B_i \stackrel{\pm}{\leftarrow} \text{tr} I_d) \Big|_{t=0} \geq \frac{d}{dt} \mathcal{L}(B_i \stackrel{\pm}{\leftarrow} \text{tr} I_d) \Big|_{t=0}.
\end{aligned}$$

562 The last inequality assumes the positivity of the term involving G_l . Otherwise, one can simply flip
563 the numerator and denominator of γ and scale the derivative of F_l instead of G_l to yield an additional
564 positive term besides the risk term to finish the proof.

565 **3. Equation (36) holds for C_i, D_i .**

566 Similarly, one can verify that eqs. (22) and (23) still hold (also eqs. (24) to (27)), and finish the proof
567 by following the same reductions as Theorem 1 with F_l and G_l . $\square$

568 B.5 Proof of Proposition 3

569 *Proof.* Let $A_l = a_l I_d, B_l = b_l I_d, C_l = c_l I_d$ and $D_l = \text{diag}(I_n \otimes D_l^1, D_l^2) + I_{n+1} \otimes D_l^3 + D_l^4 \otimes D_l^5$ for
570 $l \in [1, 2]$. Let $Z_l \in \mathbb{R}^{2d \times (3n+3)}$ be the output of the l -th attention layer, and let $X_l, Y_l \in \mathbb{R}^{d \times (3n+3)}$
571 be its first and last d rows respectively. Note that Y_l in this proof does not contain y_{test} .

572 Let $D_1^1 = \begin{pmatrix} d_x^x & 0 & d_x^y \\ 0 & 0 & 0 \\ d_y^x & 0 & d_y^y \end{pmatrix}, D_1^2 = \begin{pmatrix} s_x & 0 & s_y \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ (note that the last row of D_1^2 is masked out by M , so we
573 simply set it to 0), and $D_1^5 = \begin{pmatrix} 0 & t_x & 0 \\ 0 & 0 & 0 \\ 0 & t_y & 0 \end{pmatrix}$. We use D as an abbreviation for D_1^4 , and use $d_{i,j}$ to denote
574 the elements in D . One can verify that

$$\begin{aligned}
X_1 &= X_0 + a_1 X_0 M \left(\text{diag}(I_n \otimes D_1^1, D_1^2) + I_{n+1} \otimes D_1^3 + D_1^4 \otimes D_1^5 \right) \\
&\quad \begin{bmatrix} (1 + a_1 d_x^x) x_1 & a_1 t_x \sum_{i=1}^{n+1} d_{i,1} x_i & a_1 d_x^y x_1 \\ \vdots & \vdots & \vdots \\ (1 + a_1 d_x^x) x_n & a_1 t_x \sum_{i=1}^{n+1} d_{i,n} x_i & a_1 d_x^y x_n \\ (1 + a_1 d_x^x) x_{\text{test}} & a_1 t_x \sum_{i=1}^{n+1} d_{i,n+1} x_i & a_1 d_x^y x_{\text{test}} \end{bmatrix} \\
&= \begin{bmatrix} (1 + a_1 d_x^x) x_1 & a_1 t_x \sum_{i=1}^{n+1} d_{i,1} x_i & a_1 d_x^y x_1 \\ \vdots & \vdots & \vdots \\ (1 + a_1 d_x^x) x_n & a_1 t_x \sum_{i=1}^{n+1} d_{i,n} x_i & a_1 d_x^y x_n \\ (1 + a_1 d_x^x) x_{\text{test}} & a_1 t_x \sum_{i=1}^{n+1} d_{i,n+1} x_i & a_1 d_x^y x_{\text{test}} \end{bmatrix}.
\end{aligned}$$

575 Similarly, we have

$$\begin{aligned}
Y_1 &= Y_0 + b_1 Y_0 M \left(\text{diag}(I_n \otimes D_1^1, D_1^2) + I_{n+1} \otimes D_1^3 + D_1^4 \otimes D_1^5 \right) \\
&\quad \begin{bmatrix} b_1 d_y^x y_1 & b_1 t_y \sum_{i=1}^n d_{i,1} y_i & (1 + b_1 d_y^y) y_1 \\ \vdots & \vdots & \vdots \\ b_1 d_y^x y_n & b_1 t_y \sum_{i=1}^n d_{i,n} y_i & (1 + b_1 d_y^y) y_n \\ 0 & b_1 t_y \sum_{i=1}^n d_{i,n+1} y_i & 0 \end{bmatrix} \\
&= \begin{bmatrix} b_1 d_y^x y_1 & b_1 t_y \sum_{i=1}^n d_{i,1} y_i & (1 + b_1 d_y^y) y_1 \\ \vdots & \vdots & \vdots \\ b_1 d_y^x y_n & b_1 t_y \sum_{i=1}^n d_{i,n} y_i & (1 + b_1 d_y^y) y_n \\ 0 & b_1 t_y \sum_{i=1}^n d_{i,n+1} y_i & 0 \end{bmatrix}.
\end{aligned}$$

576 By the definition of linear attention, we can show that

$$\begin{aligned}
\text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^2) &= (Y_2)_{3n+3} = b_2 Y_1 M (c_2 X_1^\top (X_1)_{3n+3} + (D_2)_{3n+3}) \\
&= b_2 c_2 a_1 d_x^y \left(\sum_{i=1}^{3n+2} (Y_1)_i (X_1)_i^\top \right) x_{\text{test}}.
\end{aligned}$$

577 Define $\Delta X_1 = [0 \quad a_1 t_x d_{n+1,1} x_{\text{test}} \quad 0 \quad \cdots \quad 0 \quad a_1 t_x d_{n+1,n+1} x_{\text{test}} \quad 0]$, and let $\bar{X}_1 = X_1 -$
 578 ΔX_1 , then $\text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^2) = \text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^2, X_1 \leftarrow \bar{X}_1) + \text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^2, X_1 \leftarrow$
 579 $\Delta X_1)$. Let $b_1 d_y^x (1 + a_1 d_x^x) + (1 + b_1 d_y^x) a_1 d_x^x = a$, $b_1 t_y a_1 t_x = b$, $b_2 c_2 a_1 d_x^y = c$, we then have

$$\begin{aligned} \text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^2, X_1 \leftarrow \bar{X}_1) &= c \left(a \sum_{i=1}^n y_i x_i^\top + b \sum_{i=1}^{n+1} \left(\sum_{j=1}^n d_{j,i} y_j \right) \left(\sum_{j=1}^n d_{j,i} x_j^\top \right) \right) x_{\text{test}} \\ &= c \left(a \sum_{i=1}^n y_i x_i^\top + b \sum_{j=1}^n \sum_{k=1}^n \left(\sum_{i=1}^{n+1} d_{j,i} d_{k,i} \right) y_j x_k^\top \right) x_{\text{test}}, \quad (37) \\ \text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^2, X_1 \leftarrow \Delta X_1) &= bc \sum_{i=1}^{n+1} \sum_{j=1}^n d_{j,i} y_j d_{n+1,i} x_{\text{test}}^\top x_{\text{test}} \\ &= bc \sum_{j=1}^n \left(\sum_{i=1}^{n+1} d_{j,i} d_{n+1,i} \right) y_j x_{\text{test}}^\top x_{\text{test}}. \quad (38) \end{aligned}$$

580 Now consider the in-context risk,

$$\begin{aligned} \mathcal{L}(V, Q) &= \mathbb{E}_{Z_0, W} \|\text{TF}(Z_0; \{V, Q\}) + W x_{\text{test}}\|_2^2 \\ &= \mathbb{E}_{Z_0, W} \left[(\text{TF}(Z_0; \{V, Q\}) + W x_{\text{test}})^\top (\text{TF}(Z_0; \{V, Q\}) + W x_{\text{test}}) \right] \\ &= \mathbb{E}_{Z_0, W} \left[(\text{TF}(Z_0; \{V, Q\}, X_1 \leftarrow \bar{X}_1) + W x_{\text{test}})^\top (\text{TF}(Z_0; \{V, Q\}, X_1 \leftarrow \bar{X}_1) + W x_{\text{test}}) \right] \\ &\quad + 2 \mathbb{E}_{Z_0, W} [\text{TF}(Z_0; \{V, Q\}, X_1 \leftarrow \Delta X_1)^\top (\text{TF}(Z_0; \{V, Q\}, X_1 \leftarrow \bar{X}_1) + W x_{\text{test}})] \\ &\quad + \mathbb{E}_{Z_0, W} [\text{TF}(Z_0; \{V, Q\}, X_1 \leftarrow \Delta X_1)^\top \text{TF}(Z_0; \{V, Q\}, X_1 \leftarrow \Delta X_1)]. \end{aligned}$$

581 In the equation above, the 3-rd part is always positive. We then examine the second part:

$$\begin{aligned} &\mathbb{E}_{Z_0, W} [\text{TF}(Z_0; \{V, Q\}, X_1 \leftarrow \Delta X_1)^\top (\text{TF}(Z_0; \{V, Q\}, X_1 \leftarrow \bar{X}_1) + W x_{\text{test}})] \\ &= \mathbb{E}_{Z_0, W} [x_{\text{test}}^\top x_{\text{test}} v_1 x_{\text{test}} + x_{\text{test}}^\top x_{\text{test}} v_2 x_{\text{test}}] = 0, \end{aligned}$$

582 where $v_1 = bc \sum_{j=1}^n \left(\sum_{i=1}^{n+1} d_{j,i} d_{n+1,i} \right) y_j^\top c \left(a \sum_{i=1}^n y_i x_i^\top + b \sum_{j=1}^n \sum_{k=1}^n \left(\sum_{i=1}^{n+1} d_{j,i} d_{k,i} \right) y_j x_k^\top \right)$
 583 and $v_2 = bc \sum_{j=1}^n \left(\sum_{i=1}^{n+1} d_{j,i} d_{n+1,i} \right) y_j^\top W$ are independent of x_{test} . Therefore, $\mathcal{L}(V, Q)$ attains its
 584 minimum only if $\text{TF}(Z_0; \{V, Q\}, X_1 \leftarrow \Delta X_1) = 0$, implying $d_{n+1,i} = 0$ for $i \in [1, n+1]$.

585 In the following analysis, we will assume that the last row of D is 0, and let $M \in \mathbb{R}^{n \times (n+1)}$ be
 586 the first n rows of D . Additionally, we will drop the d factor in eq. (37), since its position could be
 587 substituted by a and b . We then define $\tilde{W} = a \sum_{i=1}^n y_i x_i^\top + b \sum_{j=1}^n \sum_{k=1}^n \left(\sum_{i=1}^{n+1} d_{j,i} d_{k,i} \right) y_j x_k^\top$,
 588 $\tilde{X} = [x_1 \quad \cdots \quad x_n]$ and $\tilde{Y} = [y_1 \quad \cdots \quad y_n]$. One can verify that

$$\tilde{W} = a \tilde{Y} \tilde{X}^\top + b \tilde{Y} M M^\top \tilde{X}^\top = a \tilde{W} \tilde{X} \tilde{X}^\top + b \tilde{W} \tilde{X} M M^\top \tilde{X}^\top. \quad (39)$$

589 Furthermore, the in-context risk could be expanded as

$$\begin{aligned} \mathcal{L}(V, Q) &= \mathbb{E}_{Z_0, W} \|\tilde{W} x_{\text{test}} + W x_{\text{test}}\|_2^2 = \mathbb{E}_{Z_0, W} [x_{\text{test}}^\top (\tilde{W} + W)^\top (\tilde{W} + W) x_{\text{test}}] \\ &= \mathbb{E}_{Z_0, W} [\text{tr}((\tilde{W} + W)^\top (\tilde{W} + W))] \\ &= \mathbb{E}_{Z_0, W} [\text{tr}(\tilde{W}^\top \tilde{W}) + 2 \text{tr}(W^\top \tilde{W}) + \text{tr}(W^\top W)]. \end{aligned}$$

590 We will use the identity $\mathbb{E}_X [X A X^\top X B X^\top] = (\text{tr}(A) \text{tr}(B) + \text{tr}(A B^\top) + d \text{tr}(A B)) I_d$ for any
 591 $A, B \in \mathbb{R}^{n \times n}$, which can be acquired by expanding each element and applying Isserlis' theorem. Let
 592 $T_1 = \text{tr}(M M^\top)$ and $T_2 = \text{tr}(M M^\top M M^\top)$, then

$$\mathbb{E}_{Z_0, W} [\text{tr}((a \tilde{W} \tilde{X} \tilde{X}^\top + b \tilde{W} \tilde{X} M M^\top \tilde{X}^\top)^\top (a \tilde{W} \tilde{X} \tilde{X}^\top + b \tilde{W} \tilde{X} M M^\top \tilde{X}^\top))]$$

$$\begin{aligned}
&= \mathbb{E}_{Z_0, W} [a^2 \text{tr}(XX^\top W^\top WXX^\top) + 2ab \text{tr}(XX^\top W^\top WXXMM^\top X^\top)] \\
&\quad + \mathbb{E}_{Z_0, W} [b^2 \text{tr}(XMM^\top X^\top W^\top WXXMM^\top X^\top)] \\
&= d \mathbb{E}_{Z_0} [a^2 \text{tr}(XX^\top XX^\top) + 2ab \text{tr}(XX^\top XMM^\top X^\top) + b^2 \text{tr}(XMM^\top X^\top XMM^\top X^\top)] \\
&= a^2 d^2 n(n+1+d) + 2abd^2(n+1+d)T_1 + b^2 d^2(T_1^2 + (1+d)T_2).
\end{aligned}$$

593 Simultaneously, we can verify that $\mathbb{E}_{Z_0, W}[\text{tr}(W^\top W)] = d^2$ and

$$\mathbb{E}_{Z_0, W}[\text{tr}(W^\top \widetilde{W})] = \mathbb{E}_{Z_0, W}[aW^\top WXX^\top + bW^\top WXXMM^\top X^\top] = ad^2n + bd^2T_1.$$

594 Combining the results above, we aim to find the optimal a, b, M that minimize

$$\frac{1}{d^2} \mathcal{L}(V, Q) = c_0 + c_1T_1 + c_2T_1^2 + c_3T_2,$$

595 where

$$\begin{aligned}
c_0 &= a^2n(n+1+d) + 1 + 2an, \quad c_1 = 2ab(n+1+d) + 2b, \\
c_2 &= b^2, \quad c_3 = b^2(1+d).
\end{aligned}$$

596 Since $c_3 > 0$, to minimize $\mathcal{L}(V, Q)$ we need to minimize T_2 . Given that $MM^\top$ is symmetric, we
597 denote its n eigenvalues as $\lambda_i, i \in [1, n]$. Then by Cauchy-Schwarz inequality,

$$\text{tr}(MM^\top MM^\top) = \sum_{i=1}^n \lambda_i^2 \geq \frac{1}{n} \left(\sum_{i=1}^n \lambda_i \right)^2 = \frac{1}{n} \text{tr}^2(MM^\top).$$

598 Therefore, $\mathcal{L}(V, Q)$ is minimized only if the inequality above holds with equality, which implies
599 that $\lambda_i = \lambda_j$ for any $i \neq j$. This concludes the proof by showing that there exists $\lambda \in \mathbb{R}$ such that
600 $MM^\top = \lambda I_d$, and thus $DD^\top = \text{diag}(\lambda I_d, 0)$. $\square$

601 B.6 Proof of Proposition 6

602 *Proof.* We will continue from eqs. (37) and (38). After applying token-wise dropout, we have

$$\begin{aligned}
&\text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^2, X_1 \leftarrow \bar{X}_1) = \sum_{i=1}^n (ao_2^{3i-2} + bo_2^{3i}) o_1^{3i-2} o_1^{3i} y_i x_i^\top o_1^{3n+1} o_2^{3n+3} x_{\text{test}} \\
&\quad + c \sum_{j=1}^n \sum_{k=1}^n \left(\sum_{i=1}^{n+1} o_2^{3i-1} d_{j,i} d_{k,i} \right) o_1^{3j} o_1^{3k-2} y_j x_k^\top o_1^{3n+1} o_2^{3n+3} x_{\text{test}}, \quad (40) \\
&\text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^2, X_1 \leftarrow \Delta X_1) = co_2^{3n+3} \sum_{j=1}^n \left(\sum_{i=1}^{n+1} d_{j,i} d_{n+1,i} \right) o_1^{3j} o_1^{3n+1} y_j x_{\text{test}}^\top x_{\text{test}},
\end{aligned}$$

603 where $a = b_2 c_2 a_1 d_y^y b_1 d_x^x (1 + a_1 d_x^x)$, $b = b_2 c_2 a_1 d_y^y (1 + b_1 d_y^x) a_1 d_x^x$ and $c = b_2 c_2 a_1 d_y^y b_1 t_y a_1 t_x$. One
604 can verify that our previous analysis about $\text{TF}(Z_0; \{V_l, Q_l\}_{l=1}^2, X_1 \leftarrow \Delta X_1)$ still holds and we thus
605 have $d_{n+1,:} = 0$. We then define:

$$\begin{aligned}
O_l^1 &= \text{diag}(o_l^1, \dots, o_l^{3n-2}) \in \mathbb{R}^{n \times n}, \quad O_l^2 = \text{diag}(o_l^3, \dots, o_l^{3n}) \in \mathbb{R}^{n \times n}, \quad \text{for } l \in [2], \\
O_2^3 &= \text{diag}(o_2^3, \dots, o_2^{3n+2}) \in \mathbb{R}^{(n+1) \times (n+1)}.
\end{aligned}$$

606 By defining

$$\widetilde{W} = \sum_{i=1}^n (ao_2^{3i-2} + bo_2^{3i}) o_1^{3i-2} o_1^{3i} y_i x_i^\top + c \sum_{j=1}^n \sum_{k=1}^n \left(\sum_{i=1}^{n+1} o_2^{3i-1} d_{j,i} d_{k,i} \right) o_1^{3j} o_1^{3k-2} y_j x_k^\top.$$

607 One can verify that

$$\widetilde{W} = A + B + C \triangleq aYO_1^2 O_2^1 O_1^1 X^\top + bYO_1^2 O_2^2 O_1^1 X^\top + cYO_1^2 MO_2^3 M^\top O_1^1 X^\top.$$

608 Then, we will compute the expectation of each term in the following decomposition:

$$\mathcal{L}(V, Q) = \mathbb{E}_{Z_0, W} \left[\text{tr}(\widetilde{W}^\top \widetilde{W}) + 2 \text{tr}(W^\top \widetilde{W}) + \text{tr}(W^\top W) \right],$$

609 Specifically, let $T_1 = \text{tr}(MM^\top)$, $T_2 = \text{tr}(MM^\top MM^\top)$, $T_3 = \|M\|_4^4$, $T_4 = \sum_{i=1}^n \|M_{i,:}\|_2^4$,
 610 $T_5 = \sum_{j=1}^{n+1} \|M_{:,j}\|_2^4$, we then have

$$\begin{aligned} \mathbb{E}[\text{tr}(A^\top A)] &= a^2 d^2 (np^3 + n(n-1)p^6 + (1+d)np^3), \\ \mathbb{E}[\text{tr}(B^\top B)] &= b^2 d^2 (np^3 + n(n-1)p^6 + (1+d)np^3), \\ \mathbb{E}[\text{tr}(C^\top C)] &= c^2 d^2 (p^6 T_1^2 + (1+d)(p^4 - p^6)T_4 + (1+d)(p^5 - p^6)T_5 \\ &\quad + (1+d)(p^3 - p^4 - p^5 + p^6)T_3 + (p^3 - p^4)T_4 + p^4 T_2 + dp^6 T_2), \\ \mathbb{E}[\text{tr}(A^\top B)] &= abd^2 (np^4 + n(n-1)p^6 + (1+d)np^4), \\ \mathbb{E}[\text{tr}(A^\top C)] &= acd^2 ((p^4 + (n-1)p^6)T_1 + (1+d)p^4 T_1), \\ \mathbb{E}[\text{tr}(B^\top C)] &= bcd^2 ((p^4 + (n-1)p^6)T_1 + (1+d)p^4 T_1), \\ \mathbb{E}[\text{tr}(W^\top A)] &= ad^2 np^3, \quad \mathbb{E}[\text{tr}(W^\top B)] = bd^2 np^3, \quad \mathbb{E}[\text{tr}(W^\top C)] = cd^2 p^3 T_1. \end{aligned}$$

611 Summarizing our analysis above, $\min_M \mathcal{L}(V, Q)$ is equivalent to:

$$\min_M \{c_0 + c_1 T_1 + c_2 T_2 + c_3 T_3 + c_4 T_4 + c_5 T_5 + c_6 T_1^2\},$$

612 where

$$\begin{aligned} c_0 &= 1 + n(2+d)p^3(a^2 + b^2) + 2np^3(a+b) + 2n(2+d)p^4 ab + n(n-1)p^6(a+b)^2, \\ c_1 &= 2(a+b)c(p^4 + (n-1)p^6 + (1+d)p^4) + 2cp^3, \\ c_2 &= c^2(p^4 + dp^6), \\ c_3 &= c^2(1+d)(p^3 - p^4 - p^5 + p^6), \\ c_4 &= c^2((1+d)(p^4 - p^6) + (p^3 - p^4)), \\ c_5 &= c^2(1+d)(p^5 - p^6), \\ c_6 &= c^2 p^6. \end{aligned}$$

613 It is easy to verify that $c_2, c_3, c_4, c_5, c_6 \geq 0$. □

614 B.7 Proof of Proposition 7

615 **Proposition 7 (Restate).** Let d_p denote the number of non-EOS tokens. Given any L -layer, single-
 616 head, d -dimensional linear-attention transformer with EOS tokens:

$$\text{TF}(Z_0; \{V_l, Q_l, P_l\}_{l \in [L]}) = (Z_L)_{:,d_p+1}, \quad (Z_0)_{:,d_p+1} = 0.$$

617 where

$$\begin{aligned} Z_l &\in \mathbb{R}^{d \times (d_p+1)}, \quad V_l, Q_l \in \mathbb{R}^{d \times d}, \quad P_l \in \mathbb{R}^{(d_p+1) \times (d_p+1)}, \\ Z_l &= Z_{l-1} + V_l Z_{l-1} M (Z_{l-1}^\top Q_l Z_{l-1}^\top + P_l), \quad M = \text{diag}(I_{d_p}, 0). \end{aligned}$$

618 There exists an L -layer, two-head, $2d$ -dimensional linear-attention transformer operating without
 619 EOS tokens:

$$\text{TF}(\bar{Z}_0; \{\bar{V}_l^h, \bar{Q}_l^h, \bar{P}_l^h\}_{l \in [L], h \in [2]}) = (\bar{Z}_L)_{d:2d, d_p},$$

620 where

$$\begin{aligned} \bar{Z}_l &\in \mathbb{R}^{2d \times d_p}, \quad \bar{V}_l^h, \bar{Q}_l^h \in \mathbb{R}^{2d \times 2d}, \quad \bar{P}_l^h \in \mathbb{R}^{d_p \times d_p}, \\ \bar{Z}_l &= \bar{Z}_{l-1} + \sum_{h=1}^2 \bar{V}_l^h \bar{Z}_{l-1} (\bar{Z}_{l-1}^\top \bar{Q}_l^h \bar{Z}_{l-1}^\top + \bar{P}_l^h). \end{aligned}$$

621 Such that for any $Z \in \mathbb{R}^{d \times d_p}$, by letting $Z_0 = [Z \ 0]$ and $\bar{Z}_0 = \begin{bmatrix} Z \\ 0 \end{bmatrix}$, we have

$$\text{TF}(Z_0; \{V_l, Q_l, P_l\}_{l \in [L]}) = \text{TF}(\bar{Z}_0; \{\bar{V}_l^h, \bar{Q}_l^h, \bar{P}_l^h\}_{l \in [L], h \in [2]}).$$

622 *Proof.* We construct $\bar{V}_l^h$, $\bar{Q}_l^h$, and $\bar{P}_l^h$ as follows:

$$\begin{aligned}\bar{V}_l^1 &= \begin{bmatrix} V_l & 0 \\ 0 & 0 \end{bmatrix}, \quad \bar{Q}_l^1 = \begin{bmatrix} Q_l & 0 \\ 0 & 0 \end{bmatrix}, \quad \bar{P}_l^1 = (P_l)_{1:d_p, 1:d_p}, \\ \bar{V}_l^2 &= \begin{bmatrix} 0 & 0 \\ V_l & 0 \end{bmatrix}, \quad \bar{Q}_l^2 = \begin{bmatrix} 0 & Q_l \\ 0 & 0 \end{bmatrix}, \quad \bar{P}_l^2 = \begin{bmatrix} 0 & (P_l)_{:, d_p+1} \end{bmatrix}.\end{aligned}$$

623 We will show that for any $l \in [L]$, it satisfies $\bar{Z}_l = \begin{bmatrix} (Z_l)_{:, (1:d_p-1)} & (Z_l)_{:, d_p} \\ 0 & (Z_l)_{:, d_p+1} \end{bmatrix}$. One can verify that

624 it holds trivially for $l=0$. Then, suppose it holds for some $l=k-1$, we have

$$\begin{aligned}\bar{Z}_k &= \bar{Z}_{k-1} + \bar{V}_k^1 \bar{Z}_{k-1} (\bar{Z}_{k-1}^\top \bar{Q}_k^1 \bar{Z}_{k-1} + \bar{P}_k^1) + \bar{V}_k^2 \bar{Z}_{k-1} (\bar{Z}_{k-1}^\top \bar{Q}_k^2 \bar{Z}_{k-1} + \bar{P}_k^2) \\ &= \bar{Z}_{k-1} + \begin{bmatrix} V_k (Z_{k-1})_{:, 1:d_p} \left((Z_{k-1})_{:, 1:d_p}^\top Q_k (Z_{k-1})_{:, 1:d_p} + (P_k)_{1:d_p, 1:d_p} \right) \\ 0 \end{bmatrix} \\ &\quad + \begin{bmatrix} 0 \\ V_k (Z_{k-1})_{:, 1:d_p} \end{bmatrix} \left(\begin{bmatrix} 0 & (Z_{k-1})_{:, 1:d_p}^\top Q_k (Z_{k-1})_{:, d_p+1} \end{bmatrix} + \begin{bmatrix} 0 & (P_k)_{:, d_p+1} \end{bmatrix} \right) \\ &= \bar{Z}_{k-1} + \begin{bmatrix} V_k Z_{k-1} M (Z_{k-1}^\top Q_k (Z_{k-1})_{:, 1:d_p} + (P_k)_{:, 1:d_p}) \\ 0 \end{bmatrix} \\ &\quad + \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & V_k Z_{k-1} M (Z_{k-1}^\top Q_k (Z_{k-1})_{:, d_p+1} + (P_k)_{:, d_p+1}) \end{bmatrix} \\ &= \begin{bmatrix} (Z_k)_{:, 1:d_p} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & (Z_k)_{:, d_p+1} \end{bmatrix}.\end{aligned}$$

625 The proof is complete. $\square$

626 C Experiment Details and Additional Results

627 In this section, we present experiment details and additional results not included in the main text due
628 to space limitations. Our experiments are conducted on an A100 40G GPU. It takes around 30 GPU
629 hours to fully reproduce our results¹.

630 C.1 Synthetic Experiments on Linear Transformers

631 We consider training linear attention transformers on random linear regression instances. We take
632 embedding dimension $d=4$, and the distributions for generating x_i and w_i are both $P_x = P_w =$
633 $\mathcal{N}(0, I_d)$. We optimize the ICL risk for L -layer linear transformers with n in-context demonstrations
634 using AdamW, where $L \in [3]$ and $n \in [5, 30]$. Each gradient step is computed from a batch size of
635 1000. We additionally apply ℓ_1 regularization to simplify the found solutions. For training efficiency
636 and stability, we restrict the A_l , B_l , and C_l matrices to S_l during training, and initialize $D_l \in \mathbb{R}^{d_p \times d_p}$
637 with i.i.d. Gaussian matrices. For each case, we train 40 models with different random seeds, and
638 report the minimum achieved ICL risk to approximate the global minimum.

639 To reproduce the task vector mechanism, we focus on transformers trained with triplet-formatted
640 prompts. The training procedure is identical to the above. For inference, we restrict P_u to rank-one
641 coefficient matrices, by letting $W = w_1 w_2^\top$, where $w_1, w_2 \sim \mathcal{N}(0, I_d)$. We first generate normal ICL
642 prompts to generate task vectors as the hidden states of the last arrow token after the first attention
643 layer, and then inject them into zero-shot prompts after normalization. The final outputs $\hat{y}_{\text{test}}$ are taken
644 as the output of these injected zero-shot prompts after being processed with the same transformer
645 model. We compute the final risk as $\mathbb{E} \left\| \frac{\hat{y}_{\text{test}}}{\|\hat{y}_{\text{test}}\|} + \frac{y_{\text{test}}}{\|y_{\text{test}}\|} \right\|$ to simulate the layer normalization blocks
646 in practical LLMs. The reported scores are averaged for $n \in [5, 30]$.

647 C.2 Experiments on Practical LLMs

648 **Datasets.** Following the settings of the original task vector method [7], our study covers 33 tasks in 5
649 categories. The detailed description for each task is provided in Table 3.

¹The source code is available in supplementary materials.

Table 3: Descriptions of the tasks used in our empirical studies.

Category	Task	Example	Description
Knowledge	Contry to Capital	France → Paris	Output the capital city of the given country.
	Person to Language	Macron → French	Output the native language of the given person.
	Location to Continent	Paris → Europe	Output the corresponding continent of the given location.
	Religion	Saladin → Muslim	Output the associated religion of the given location or person.
Algorithmic	List First	[a,b,c] → a	Output the first item in the given list.
	List Last	[a,b,c] → c	Output the last item in the given list.
	Next Letter	a → b	Output the next letter of the given letter in the alphabet.
	Prev Letter	b → a	Output the previous letter of the given letter in the alphabet.
	To Upper	a → A	Output the corresponding uppercase letter of the given lowercase letter.
	To Lower	A → a	Output the corresponding lowercase letter of the given uppercase letter.
Translation	English to French	hello → bonjour	Translate the given word in English to French.
	English to Italian	hello → ciao	Translate the given word in English to Italian.
	English to Spanish	hello → hola	Translate the given word in English to Spanish.
	French to English	bonjour → hello	Translate the given word in French to English.
	Italian to English	ciao → hello	Translate the given word in Italian to English.
	Spanish to English	hola → hello	Translate the given word in Spanish to English.
Linguistic	Present to Gerund	go → going	Output the corresponding gerund form of the given verb in present simple tense.
	Present to Past	go → went	Output the corresponding past simple form of the given verb in present simple tense.
	Present to Past Perfect	go → gone	Output the corresponding past perfect form of the given verb in present simple tense.
	Gerund to Present	going → go	Output the corresponding present simple form of the given verb in gerund form.
	Past to Present	went → go	Output the corresponding present simple form of the given verb in past simple tense.
	Past Perfect to Present	gone → go	Output the corresponding present simple form of the given verb in past perfect tense.
	Singular to Plural	dog → dogs	Output the corresponding plural form of the given noun in singular form.
	Plural to Singular	dogs → dog	Output the corresponding singular form of the given noun in plural form.
	Antonym	happy → sad	Output the antonym of the given adjective.
Bijection	To Upper & Lower	a ↔ A	Output the given letter in uppercase if it is in lowercase, and vice versa.
	English & French	hello ↔ bonjour	Translate the given word to French if it is in English, and vice versa.
	English & Italian	hello ↔ ciao	Translate the given word to Italian if it is in English, and vice versa.
	English & Spanish	hello ↔ hola	Translate the given word to Spanish if it is in English, and vice versa.
	Present & Gerund	go ↔ going	Output the given verb in gerund form if it is in present simple tense, and vice versa.
	Present & Past	go ↔ went	Output the given verb in past simple form if it is in present simple tense, and vice versa.
	Present & Past Perfect	go ↔ gone	Output the given verb in past perfect form if it is in present simple tense, and vice versa.
	Singular & Plural	dog ↔ dogs	Output the given noun in plural form if it is in singular form, and vice versa.

Table 4: Accuracy comparison between standard ICL (Baseline), the task vector method (TaskV), and our strategy (TaskV-M). The experiment is conducted on Pythia-12B with $n = 10$.

	Method	Knowledge	Algorithmic	Translation	Linguistic	Bijection	Average
0-shot	Baseline	6.60 $\pm$ 1.59	14.07 $\pm$ 1.45	8.60 $\pm$ 0.68	12.53 $\pm$ 1.57	10.31 $\pm$ 0.70	10.82 $\pm$ 0.48
	TaskV	63.30 $\pm$ 2.62	84.73 $\pm$ 1.22	62.07 $\pm$ 0.98	82.58 $\pm$ 1.22	42.27 $\pm$ 0.92	66.40 $\pm$ 0.96
1-shot	Baseline	61.80 $\pm$ 5.45	72.80 $\pm$ 1.15	43.27 $\pm$ 2.92	57.07 $\pm$ 1.15	41.91 $\pm$ 2.83	53.95 $\pm$ 1.02
	TaskV	76.40 $\pm$ 2.40	84.20 $\pm$ 1.05	71.47 $\pm$ 1.41	87.16 $\pm$ 2.04	53.11 $\pm$ 2.37	73.59 $\pm$ 0.79
	TaskV-M	77.70 $\pm$ 2.52	83.73 $\pm$ 1.37	71.00 $\pm$ 1.48	86.80 $\pm$ 1.59	53.87 $\pm$ 2.90	73.68 $\pm$ 0.90
2-shot	Baseline	70.30 $\pm$ 3.71	82.13 $\pm$ 0.54	60.80 $\pm$ 1.81	81.16 $\pm$ 1.57	50.76 $\pm$ 2.17	68.41 $\pm$ 0.64
	TaskV	80.30 $\pm$ 2.46	87.00 $\pm$ 1.63	76.13 $\pm$ 3.77	89.33 $\pm$ 0.70	58.67 $\pm$ 2.44	77.41 $\pm$ 0.50
	TaskV-M	81.60 $\pm$ 1.56	86.47 $\pm$ 0.40	77.27 $\pm$ 2.53	89.51 $\pm$ 0.88	59.24 $\pm$ 2.48	77.87 $\pm$ 0.76
3-shot	Baseline	77.60 $\pm$ 2.40	81.87 $\pm$ 0.81	68.13 $\pm$ 2.02	86.31 $\pm$ 1.93	55.73 $\pm$ 1.60	73.20 $\pm$ 0.31
	TaskV	84.00 $\pm$ 2.76	86.33 $\pm$ 1.17	79.53 $\pm$ 2.27	92.00 $\pm$ 0.67	58.76 $\pm$ 1.53	79.06 $\pm$ 0.67
	TaskV-M	85.40 $\pm$ 2.31	87.07 $\pm$ 1.18	78.13 $\pm$ 1.86	92.84 $\pm$ 0.68	59.56 $\pm$ 1.27	79.54 $\pm$ 0.35
4-shot	Baseline	78.40 $\pm$ 1.83	82.73 $\pm$ 0.44	72.40 $\pm$ 1.24	88.89 $\pm$ 1.25	57.91 $\pm$ 1.46	75.46 $\pm$ 0.64
	TaskV	83.80 $\pm$ 1.12	87.60 $\pm$ 1.81	80.20 $\pm$ 2.39	92.18 $\pm$ 0.96	59.38 $\pm$ 0.47	79.59 $\pm$ 0.62
	TaskV-M	84.30 $\pm$ 1.50	88.13 $\pm$ 0.81	80.00 $\pm$ 2.67	91.87 $\pm$ 1.25	60.31 $\pm$ 0.86	79.87 $\pm$ 0.51

Prompt Template. The template used to construct ICL demonstrations is “Example: $\{x_i\} \rightarrow \{y_i\}$, where x_i and y_i are subsequently replaced by the input and output of the semantic mapping. For the query part, y_i is omitted from the prompt. After concatenating each demonstration with “\n”, an example of the full input prompt is:

Example: $\{x_1\} \rightarrow \{y_1\}$ \n \cdots Example: $\{x_n\} \rightarrow \{y_n\}$ \n Example: $\{x_{\text{test}}\} \rightarrow$

Evaluation. To evaluate the N -shot performance, we generate $50 \times (N + 1)$ i.i.d. prompts for each task with number of demonstrations $n = 10$ for task vector extraction. The hidden states of the last $\Rightarrow$ token, which is also literally the last token in the prompt, are recorded for every layer in the transformer. Thereafter, we generate another 50 i.i.d. prompts with N demonstrations, where x_{test} is selected to be distinct from the previous chosen ones. The final accuracy is measured by whether the next word predicted matches the expected answer. The performance of the standard ICL method (Baseline) is acquired by inferring without interference. For the task vector method (TaskV) and our multi-vector variant (TaskV-M), the extracted task vectors are injected to replace the hidden states of the arrow $\Rightarrow$ tokens at a specified layer l . For TaskV, only the last arrow token is injected, while for TaskV-M, each of the $N + 1$ arrow tokens is injected with the $N + 1$ extracted task vectors for the same task. The performance is reported for the one layer $l \in L$ achieving the highest accuracy. For each case, the mean and standard deviation are evaluated through 5 independent trials.

Additional Results. Besides Llama-13B, we also observe consistent accuracy improvement of our TaskV-M method on the Pythia-12B model, as reported in Table 4.

D Additional Discussions

D.1 Last Task Vector Weights the Most

While our analysis of linear-attention models suggests that each formed task vector (i.e., the hidden state at each arrow token) contributes equally to the final prediction, this assumption does not fully hold in practical LLMs. As demonstrated by the conflicting tasks experiment in [7], injecting a task vector from task B into an ICL prompt designed for task A causes the model to predominantly perform task B . This behavior indicates that LLMs largely rely on the last arrow token to determine the task identity. We attribute this to the causal attention mechanism used in practical LLMs, which is not captured by our current theoretical analysis. In causal attention, only the final arrow token can aggregate information from the entire preceding context, making it the most informative and influential for prediction. This explains why our multi-vector strategy offers modest, though consistent, performance gains. The improvement suggests that intermediate arrow tokens do participate in the inference process, albeit less effectively. Enhancing how LLMs utilize information from all arrow tokens remains a promising direction for improving task vector accuracy and robustness.

682 D.2 Decoding the Vocabulary of Task Vectors

683 Multiple prior works [7, 16] have observed an interesting phenomenon: when task vectors are
684 extracted and passed through the final classification layer, the top predicted tokens often belong to
685 the output space of the corresponding task. This effect is particularly prominent in the GPT-J model.
686 Interestingly, we find that this behavior can be naturally explained by our analysis of linear models.
687 Specifically, we assume that the hidden state space has dimensionality at least $2d$, where the first d
688 dimensions represent the input (x_i) and the last d dimensions represent the output (y_i). Task vectors
689 constructed under this architecture preserve this layout: the first half encodes a linear combination
690 of x_i , and the second half encodes a linear combination of y_i . In the final layer, the model predicts
691 y_{test} by extracting the last d dimensions of the final token. When this same mechanism is applied to
692 a task vector, it naturally produces a linear combination of the y_i values, thereby generating outputs
693 aligned with the task’s output space. This indicates that practical LLMs adopt a similar partition in
694 the hidden state space, justifying our prompt structure for linear model analysis.

695 D.3 Limitations

696 While our analysis provides new insights into the emergence and functionality of task vectors, it is
697 primarily conducted on simplified linear-attention transformers and synthetic tasks, which may not
698 fully capture the complexity of real-world LLMs. Moreover, our theoretical framework focuses on
699 middle-layer representations and does not fully account for deeper interactions across layers or the
700 role of fine-tuned components such as layer normalization and multi-head attention.

701 D.4 Broader Impacts

702 This work advances the theoretical understanding of in-context learning and task vector mechanisms,
703 which can lead to more efficient and interpretable language models. By enabling faster inference
704 through task vectors, it may reduce the computational cost and energy consumption of large-scale
705 deployment, thereby making AI systems more accessible and environmentally sustainable. Im-
706 proved interpretability could also enhance trust and transparency in AI applications across education,
707 healthcare, and other socially beneficial domains.

708 As task vector methods improve efficiency and transferability, they may also be misused to replicate
709 or extract functionality from proprietary models without authorization, raising concerns around model
710 intellectual property. Additionally, while interpretability is often framed as a benefit, deeper insights
711 into model internals could be exploited to engineer adversarial inputs or extract sensitive training
712 data. Careful consideration and mitigation strategies are essential to ensure that such work aligns
713 with the broader goals of safe and beneficial AI.

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